Supplemental Material for "Role differentiation as ignition of a collective information engine"

Maximilian Puelma Touzel[1,*]

[1] *Mila–Quebec AI Institute, 6666 St. Urbain Ave., Montreal, Quebec, Canada*

(Dated: November 2, 2026)

CONTENTS

Notation 3

Notation 3

S1. The model ladder and the coordination payoff 7

A. The ladder of six models 7

B. The coordination payoff 9

C. Role-following as a self-enforcing correlated equilibrium 10

D. Learning the policy and selecting the convention 11

S2. Pairing statistics and the engine parameter 14

A. Beyond mean field: pairwise error rates 14

B. Pairing efficiency as a partner correlation 14

C. The three factors 18

D. Pairing as a control on the cascade class 20

S3. Role capacity and associative-memory structure 21

S4. Where the ω-identity is kept 22

A. Repertoire seeding and the ledger's memory 24

S5. Monotonicity of the pinning fixed point for general repertoires 25

S6. Validity of the linearization and the fluctuation-shifted threshold 26

S7. Scope of the regime map 28

S8. From the encounter process to the mean-field drift 29

S9. Sketch of an entropy and information ledger 33

S10. Simulation details 39

References 39

* puelmatm@mila.quebec

NOTATION

Symbols are grouped so that quantities separated by a single averaging step, or by one stage of the loop, sit together. Equation numbers without an S prefix refer to the main text.

Status and its differences. Distinguished by which partner information has been averaged away.

x_i	Projection of agent i in context m, $\boldsymbol{\omega}_i \cdot \mathbf{g}_m$.
$\tanh x_i$	Status of agent i in context m, the score; what agent i presents.
$\Delta_{im}(j)$	Difference to a named partner, $\tanh x_i - \tanh x_j$; what one encounter presents.
Δ_{im}	After the partner average, $\tanh x_i - \mathbb{E}[\tanh x_j \mid x_i]$, near the fixed point $(1-\rho_{\text{pair}})x_i$; what enters Eq. (2).
Δ	Pairwise difference at fixed context and partner (Appendix A, Sec. S8).
s_{im}	Observed signal, $\Delta_{im}(j) + \sigma_{\text{read}}\xi$; the only status an agent sees.

Action and policy. Distinguished by which noise has been averaged away.

a	Action, -1 (Defer) or $+1$ (Go).
T	Evidence samples per decision; the read memory.
π	Action policy, $p(a \mid \{s_{im,t}\}_{t=1}^{T})$.
$\bar{\pi}(\Delta)$	Go-probability at true difference Δ; the policy averaged over read noise.
$\bar{a}(\Delta)$	Noise-averaged action at a *given* partner, $2\bar{\pi}(\Delta) - 1$; odd in Δ.
$\langle a \rangle_m(\boldsymbol{\omega}_i)$	Noise- *and* partner-averaged action; the response in Eq. (2).
ℓ	Slope of the learnable policy π_ℓ (Sec. S1 D); $\ell \to \infty$ the sign rule, $\ell < 0$ anti-follow.

Gains. One per stage, one for the closed loop.

g	Read gain, $2\bar{\pi}'(0) = \sqrt{2T/\pi}/\sigma_{\text{read}}$; slope of $\bar{a}$ at zero difference.
η_{agent}	Inference efficiency, $\sigma_{\text{read}} g/2$; equals $\sqrt{T/2\pi}$ for T samples.
$g(\ell)$, $\ell_c^{(k)}$	Read gain of π_ℓ, $\to g$ as $\ell \to \infty$; slope at which axis k ignites.
$\mu_{\text{eff}}(v)$	Read gain softened by partner variance v; $\to g$ as $v \to 0$.
gu_m	Axis m's share of the loop gain: read times write, each $\propto \sqrt{u_m}$.
$\boldsymbol{\Lambda}$	Social loop gain matrix, Eq. (5).
Λ_k	Its k-th eigenvalue; axis k ignites at $\Lambda_k = 1$.

Noise. Two independent sources, and what the reduction discards.

σ_{read}	Observation (read) noise; the finest status difference resolved.
σ_ω	ω-identity noise; variation in $\boldsymbol{\omega}_i$ from outside the modeled encounters.
ξ, $\boldsymbol{\eta}_i$	Standard Gaussian sources for the two noises above.
v	Partner-projection variance $\mathbf{C}_{mm}$, of order σ_ω^2 subcritically.
$\beta b_0 = \beta^2/\nu$	Shot-noise intensity discarded by the dense-encounter limit.

State and its spectra.

$\boldsymbol{\omega}_i$	ω-identity of agent i, a vector in $\mathbb{R}^d$.
$\boldsymbol{\Omega}$	The N_{a} ω-identities stacked.
r_i	Accumulated coordination income (wealth) of agent i.
$\mathbf{g}_m$, $\hat{\mathbf{g}}_m$	Schema vector and its direction, $\mathbf{g}_m = \sqrt{u_m}\,\hat{\mathbf{g}}_m$.
u_m	Schema strength $\lvert\mathbf{g}_m\rvert^2$; the power axis m puts on the channel.
$\boldsymbol{H}$	The $M \times d$ matrix of stacked schema vectors.
$\mathbf{C}$	Population covariance $N_{\text{a}}^{-1}\sum_i \boldsymbol{\omega}_i\boldsymbol{\omega}_i^\top$; the order parameter.
$\mathbf{C}_r$	Resource-weighted covariance, $\sum_i r_i\boldsymbol{\omega}_i\boldsymbol{\omega}_i^\top / \sum_i r_i$.
v_k	k-th eigenvalue of $\mathbf{C}$; leaves the floor $\sigma_\omega^2/2\gamma$ when k ignites.
v_m	Variance of $\mathbf{C}$ along $\hat{\mathbf{g}}_m$, $\hat{\mathbf{g}}_m^\top\mathbf{C}\hat{\mathbf{g}}_m$; equals v_k for an orthogonal repertoire.
μ_k	k-th eigenvalue of $\boldsymbol{H}^\top\boldsymbol{H}$; equals u_k for an orthogonal repertoire.
λ_m	Resource-weighted variance along $\hat{\mathbf{g}}_m$, $\hat{\mathbf{g}}_m^\top\mathbf{C}_r\,\hat{\mathbf{g}}_m$.

Rates.

γ	ω-identity decay rate; $1/\gamma$ is the write memory.
ν	Encounter rate per agent, a fixed budget independent of N_{a}.
b_0	Per-encounter kick to the ω-identity.
$\beta = \nu b_0$	Feedback strength; the rate decisions are written into ω-identity.
c	Resource decay rate; $1/c$ the income horizon.
α, κ	Schema reinforcement and decay rates.
$\varphi(U)$	Budget pressure of the shared channel, common to every schema.
r_{comp}, r_{act}	Compounding and activation rates of the u-transient; $r = r_{\text{act}}/r_{\text{comp}}$.
τ_k	Relaxation time of mode k, $1/\gamma\lvert1-\Lambda_k\rvert$.

Population and pairing. Three different counts of agents.

N_{a}	Number of agents.
N_{enc}	Encounters in one coarse-graining window, $\nu\Delta t$ (Sec. S8).
N_{eff}	Participation ratio of the pairing kernel; $N_{\mathrm{a}} - 1$ if uniform.
f_m	Play frequency of context m, with $\sum_m f_m = 1$.
$p_{ij}^{(m)}$, $\boldsymbol{P}^{(m)}$	Pairing probabilities set by the matching rule, and their kernel.
ρ_{pair}	Correlation of a partner's projection with one's own, set by that rule.
$\boldsymbol{L}^{(m)}$	Graph Laplacian $\mathbf{I} - \boldsymbol{P}^{(m)}$; how a persistent network would enter.

Repertoire and capacity. Four counts of schemas, by what limits each.

M, d	Number of contexts and embedding dimension.
$q = M/d$	Aspect ratio of the repertoire.
R	$\mathrm{rank}(\boldsymbol{H}) \leq \min(M, d)$.
u_0, σ_u	Seed scale and log-dispersion, $u_m(0) = u_0 e^{\sigma_u z_m}$.
u_{cap}	Per-axis range ceiling of the channel.
$U = \sum_m u_m$	Total channel power, limited by the capacity U_{cap}.
M_{cap}	$U_{\mathrm{cap}}/u_{\mathrm{cap}}$; axes drivable to full range at once.
Θ	Capacity ratio $(U_{\mathrm{cap}}/M)/u_{\mathrm{cap}} = M_{\mathrm{cap}}/M$; share per schema in ceiling units.
M_{eff}	Axes exceeding the schema decay rate at a self-consistent fixed point, $\lambda_k(\mathbf{C}_r) \geq \kappa/\alpha$.
K	Realized survivor count at the end of the u-transient; need not saturate.
$N(\Lambda)$	Thickness; active role axes at loop gain Λ.

Payoff and information.

w_0	Maximal coordination gain per encounter.
W_{ijm}, W_{im}	Pairwise payoff above Nash and its partner average.
$P_{e,ijm}$, $P_{e,im}$	Pairwise and partner-averaged role-inference error rates.
I	Role information $1 - H_{\mathrm{bin}}(P_e)$, which bounds the payoff.
c_{col}, Δ_0	Status tilt of the collision payoff and its scale (Sec. S1 D).
$\bar{r}_{\mathrm{rf}}$	Role-free resource mean, of order σ_ω^2.

Terms. Defined terms without a symbol of their own; those with one are glossed in the tables above.

Structuration.

structuration	Giddens's resolution of that debate: structure as medium and outcome.
duality of structure	Structure is both the medium and the outcome of what it organizes.
structure–agency debate	Whether structure shapes individuals or their choices generate it.
schemas	Transposable rules and conventions that prescribe behavior around resources.
resources	Material and symbolic stocks that schemas mobilize and that empower users.
transposability	That a schema carries across contexts; here the overlap $\hat{\mathbf{g}}_m \cdot \hat{\mathbf{g}}_n$.
institutionalization	A schema gaining strength; allocation of channel capacity across games.

Engine and channel.

status channel	The collective self-sensing process by which agents read status differences.
status	Label for the score $\tanh(\boldsymbol{\omega}_i \cdot \mathbf{g}_m)$; its reward asymmetry is immaterial to the theory.
ignition	Onset of the instability at $\boldsymbol{\omega} = 0$; roles then sustain themselves.
thick phase	Above ignition, $N \geq 1$; at least one role axis has split.
irreducible cognitive floor	$\eta_{\text{agent}} > \sigma_{\text{read}}\gamma/4\beta$; below it no matching rule sustains roles.
well-mixed	Uniform pairing, $p_{ij}^{(m)} = 1/N_{\text{a}}$; the Laplacian becomes a projector.
platform-owned	Set by the operator rather than by the agents; all factors but η_{agent}.

Repertoire dynamics.

strength sector	The u_m half of the polar split of the schema dynamics; solved here.
transformation sector	The $\hat{\mathbf{g}}_m$ half, rotating what a schema is about; left exogenous.
u-transient	Institutionalizes the repertoire: reduction to survivors, then shaping.
active set	$\{m : u_m > u_0\}$, the schemas above the seed scale; its size is K.
invasion threshold	$\alpha\lambda_m = \kappa + \varphi$, above which a schema's strength grows.
slack band	$\Theta \geq 1$, where the per-schema ceiling is reached first and all M survive.
budget-limited band	$1/M \leq \Theta < 1$, where the budget sustains M_{cap} of the M.
winner-take-all band	$\Theta < 1/M$, where the budget cannot sustain even one at the ceiling.
saturated cluster	Endpoint with the survivors together at the ceiling u_{cap}.
condensation	Runaway of all strength onto one schema when the budget cannot hold it.
pinning	Arrest of that runaway at the ceiling, with a fringe at the threshold.

Cascade classes. The five spectral shapes of $\boldsymbol{H}^\top \boldsymbol{H}$.

degenerate	Equal eigenvalues; one avalanche, with no advance warning.
geometric	Constant ratios; a logarithmic cascade, the most monitorable class.
Zipf	Power-law strengths; decelerating for $a > 1$, accelerating for $a < 1$.
spiked	One outlier over a bulk, from coherent schemas sharing axes.
random	Marčenko–Pastur bulk; quiet, then a burst at the upper edge.

Matching rules and memory modes.

random pairing	Partner drawn independently of status, $\rho_{\text{pair}} = 0$; the reference case.
rank-ordered pairing	Each rank paired with its mirror, $\rho_{\text{pair}} \to -1$; halves onset.
assortative pairing	Like paired with like, $\rho_{\text{pair}} \to 1$; the gain factor vanishes.
feature mode	Graded projections near the bifurcation, soft and high-dimensional.
prototype mode	Sharp, near-saturated projections deep in the thick phase.
bimodal split	The population occupying complementary basins rather than one.

S1. THE MODEL LADDER AND THE COORDINATION PAYOFF

A. The ladder of six models

The main text presents the closed loop directly. It was developed as a sequence of six models of increasing structure, each adding one component and exposing one phenomenon. We summarize them here (Table S1). The reductions identify which component produces which result.

Model 1 places a single agent in a Go/Defer game against a fixed environment. An oracle supplies a noisy role signal $s = \Delta + \sigma_{\text{read}}\xi$, with ξ a standard Gaussian noise source. The agent's action policy $\pi : \mathbb{R} \to [0, 1]$ maps signal to Go-probability. A policy that acts on the sign of a single observation has slope $\bar{\pi}'(0) = 1/(\sigma_{\text{read}}\sqrt{2\pi})$; pooling T signals before thresholding gives $\bar{\pi}'(0) = \sqrt{T}/(\sigma_{\text{read}}\sqrt{2\pi})$. The model has the structure of a Szilard engine: the oracle is the measurement, the policy the feedback protocol, and the coordination gain plays the role of work. The fuel is $I(\text{sign}(\Delta); s) = 1 - H_{\text{bin}}(P_e)$ bits and the work is $W = w_0(1 - 2P_e)^2$, derived in Sec. S1 B. The two satisfy $W \leq 2 \ln 2 \, w_0 \, I$ (equivalently $W \leq 2w_0 I$ with I in nats), so a coordination payoff requires role information. The bound is saturated as $P_e \to \frac{1}{2}$, and both W and I vanish in the symmetric (undifferentiated) state.

TABLE S1. The six models. Number of agents (N_a) and number of contexts (M). Each adds one component (bold) and yields one key result. Model 1 has exogenous feedback; the rest are endogenous.

#	N_a/ M	New component	Key result
1	1 / 1, ex.	policy π, external oracle	classical info engine; $\bar{\pi}'(0)$ is measurement quality
2	2 / 1,	**reciprocal measurement**	self-generated fuel; $G = 2g$, $\beta_c = \gamma/2g$
3	N_a / 1	**population, mean field**	phase transition; $G = g$; order parameter C
4	N_a / M	**vector roles**	$\mathbf{\Lambda} = \Lambda \boldsymbol{H}^\top \boldsymbol{H}$; conditional cascade
5	N_a / M	**dynamic schemas**	self-consistency; but schemas collapse to one axis
6	N_a / M	**resources**	replicator on fixed axes; endogenous cascade

Ref. [1] gives the corresponding bound on consensus (flocking) order in informational active matter. Our bound is Pinsker's inequality for the binary relative entropy, $I \geq 2(\frac{1}{2} - P_e)^2$ in nats, multiplied by $2w_0$. It does not follow from the feedback second law of Sagawa and Ueda [2] and carries no thermodynamic content.

Model 2 removes the oracle: two agents with projections x_1, x_2 each measure the other, so the fuel is self-generated. Near threshold the status difference is $\Delta \approx x_1 - x_2$, and it obeys $\dot{\Delta} \approx (-\gamma + 2\beta g)\Delta + \sqrt{2}\,\sigma_\omega \eta$, with gain $G = 2g$ (a factor 2 from the $\pm\beta$ swing, another from reciprocal amplification) and threshold $\beta_c = \gamma/2g$. For $\beta < \beta_c$ the difference decays to zero. For $\beta > \beta_c$ it grows until the reads become error-free, where the bounded write balances the decay.

Model 3 takes N_a agents randomly paired, in the mean field. Only the focal agent's update contributes, halving the gain to $G_\mathrm{MF} = g$, so $\beta_c^\mathrm{MF} = \gamma/g = \gamma\sigma_\mathrm{read}\sqrt{2\pi}/2$. The mean $\langle x \rangle$ vanishes in the thick phase, so the order parameter is the population variance $C = N_\mathrm{a}^{-1} \sum_i x_i^2$, with $C - C_0 \propto (\beta - \beta_c)$ near threshold.

Model 4 gives agents vector positions $\boldsymbol{\omega}_i \in \mathbb{R}^d$ and M schemas $\mathbf{g}_m \in \mathbb{R}^d$. The loop gain becomes a matrix on the schema second moment, $\mathbf{\Lambda} = \Lambda \sum_m \mathbf{g}_m \mathbf{g}_m^\top = \Lambda \boldsymbol{H}^\top \boldsymbol{H}$, with $\boldsymbol{H}$ the $M \times d$ matrix of stacked schemas (Sec. II A of the main text). Its eigenvalues give the cascade $\beta_c^{(k)} = \gamma/\mu_k$. The order parameter generalizes to the covariance $\mathbf{C} = N_\mathrm{a}^{-1} \sum_i \boldsymbol{\omega}_i \boldsymbol{\omega}_i^\top$, with the number of supra-floor eigenvalues—the cascade height $N(\Lambda)$—the thickness.

Model 5 lets the schema axes evolve, $\dot{\mathbf{g}}_m = -\kappa \mathbf{g}_m + \alpha \mathbf{C} \mathbf{g}_m$: a power iteration rotating

$\mathbf{g}_m$ toward the leading eigenvector of $\mathbf{C}$. Schemas align with the role structure and the role structure reinforces the schemas, which is one half of Sewell's duality. All schemas converge on the same axis, so the stationary structure is one-dimensional. The same collapse occurs for the resource-weighted rotation of Model 6, and is analyzed with its two candidate remedies in Appendix C of the main text.

Model 6 closes the loop with resources (Sec. II B of the main text): schemas are reinforced by the resource-weighted covariance $\mathbf{C}_r$, so agents who profit from a schema reinforce it. Reinforcement is thereby contingent on success rather than on raw differentiation, and schema competition becomes a replicator dynamics whose fitness is resource accumulation. The schema axes are held fixed (Sec. I of the main text), so what the loop selects is the strength of each schema, not its axis.

B. The coordination payoff

The oracle (or, endogenously, the population) assigns a leader (should Go) and follower (should Defer). Each agent's signal has independent role-inference error rate P_e. The four joint outcomes and their payoffs (averaged over the pair) are: both correct, prob $(1-P_e)^2$, payoffs $(6,2)$, average 4; leader wrong only, prob $P_e(1-P_e)$, both Defer $(3,3)$, average 3; follower wrong only, prob $(1-P_e)P_e$, both Go $(-1,-1)$, average -1; both wrong, prob P_e^2, roles swap to $(2,6)$, average 4—still coordinated. The expected average payoff is

$$\bar{U} = 4(1-P_e)^2 + 3P_e(1-P_e) - (1-P_e)P_e + 4P_e^2 = 4 - 6P_e(1-P_e). \tag{S1}$$

The symmetric mixed equilibrium has each player Go with probability p^* fixed by indifference, $-p^* + 6(1-p^*) = 2p^* + 3(1-p^*)$, so $p^* = \frac{1}{2}$ and both actions pay $U_{\text{Nash}} = \frac{5}{2}$. Subtracting it and using $6P_e(1-P_e) = \frac{3}{2} - \frac{3}{2}(1-2P_e)^2$,

$$W = \bar{U} - U_{\text{Nash}} = w_0\,(1-2P_e)^2, \qquad w_0 = \text{CE} - \text{Nash} = 1.5. \tag{S2}$$

The quadratic form is specific to two-role anti-coordination: a double error swaps the roles and leaves the pair coordinated (the fourth outcome has the same payoff as the first), so the work depends only on $(1-2P_e)^2$. Because W is quadratic in P_e, the resource income of Sec. II B of the main text is quadratic in ω-identity, which allows the Gaussian fixed-point calculation to close.

C. Role-following as a self-enforcing correlated equilibrium

The role signal acts as a correlating device [3, 4]. At each encounter the noisy status difference $s = \Delta_{im}(j) + \sigma_{\text{read}}\xi$ recommends the action $a_i = \text{sign}(s)$ to agent i. The obedience constraint is that following this recommendation is a best response when the partner also follows. Writing $q = \Pr(\text{partner plays Go} \mid s_i)$, the anti-coordination payoffs give expected returns

$$\text{Go: } 6(1-q) - q = 6 - 7q, \qquad \text{Defer: } 2q + 3(1-q) = 3 - q, \tag{S3}$$

so a Go recommendation is obedient for $6 - 7q \geq 3 - q$, that is $q \leq \frac{1}{2}$, and a Defer recommendation for $q \geq \frac{1}{2}$. The status difference is antisymmetric: an agent recommended Go is the higher-status party, whose partner is the more likely to Defer. Both constraints therefore reduce to the same requirement, that the recommendation anti-correlate with the partner's action. This is a condition on measurement quality, set by $\Delta/\sigma_{\text{read}}$ and the integration T, the same quantities that enter η_{agent}.

The equilibrium is not in dominant strategies. Following is a best response only if the partner also follows: against a partner who ignores the signal and always plays Go, the best response is to always Defer. Role-following therefore requires shared adoption. Replicator dynamics on strategies, that is imitation of higher-paying play, selects the correlated equilibrium in social dilemmas [5]. Here the correlating device is produced by the actions themselves, and both orientations of the convention, following the signal and reversing it, are self-enforcing. Selection on payoff therefore does not by itself settle which is adopted. Payoff learning reaches a role-following policy from the mixed equilibrium, and any asymmetry through which status enters the payoffs fixes its orientation (Sec. S1 D). Because the correlating device is noisy, the applicable solution concept is correlated equilibrium of the Bayesian game [6]. The policy is not a further assumption. The payoff depends on it only through the error rate, $W = w_0(1-2P_e)^2$, and P_e is smallest when an agent reports the sign of its accumulated evidence. Lowering P_e raises $\bar{\pi}'(0)$, and with it η_{agent}. Against a partner who also follows, the policy used here is therefore the one any algorithm that maximizes reward would find. Such an algorithm also reaches it from the mixed equilibrium when status bears on the outcome of a collision (Sec. S1 D). This holds where the cost of the memory supporting the inference is negligible against the payoff it yields. Once that cost is counted, the strategy that minimizes P_e need not maximize net output [7].

The obedience constraint tightens as roles differentiate. At the symmetric fixed point ($\Delta \to 0$) the signal carries no information, $q = \frac{1}{2}$, and both constraints hold with equality. The correlated equilibrium then coincides with the mixed Nash equilibrium and pays 2.5. As Δ grows the signal becomes informative, $q < \frac{1}{2}$, and the role-signal equilibrium pays 4 and is strictly self-enforcing. Ignition ($\Lambda > 1$) is the instability that takes the population from the first case to the second: the order parameter and the obedience slack $\frac{1}{2} - q$ grow together, both set by the informativeness of the signal. The correlating device here is the agents' own accumulated status rather than an external mediator.

The calculation is carried out at the symmetric fixed point, where the asymmetry between the two roles (payoffs 6 and 2) averages out. Retaining it splits the obedience constraint between the two roles, and the two conditions need not then coincide: an agent assigned the lower-paying role may prefer a less informative signal. The learners of Sec. S1 D do not test this incentive, since one shared slope serves both roles. Incentive conflict between roles enters with the stratification extension noted in the main text and is not treated here.

D. Learning the policy and selecting the convention

The sign rule need not be assumed. We embed it in a one-parameter family of single-observation ($T = 1$) policies, giving the Go-probability at signal s and its read gain,

$$\pi_\ell(s) = \tfrac{1}{2}\big[1 + \tanh(\ell s)\big], \qquad g(\ell) = 2\bar{\pi}'_\ell(0) = \ell\, \mathbb{E}_\xi\big[\operatorname{sech}^2(\ell \sigma_{\text{read}} \xi)\big]. \tag{S4}$$

At $\ell = 0$ each agent plays Go with probability $\frac{1}{2}$, the mixed Nash equilibrium of Sec. S1 B. As $\ell \to \infty$ the policy becomes the sign rule, and $g(\ell)$ rises to the sign-rule read gain $g = \sqrt{2/\pi}/\sigma_{\text{read}}$. Negative ℓ reverses the rule, so that the lower-status agent plays Go. The slope enters the loop gain only through the read gain, $\Lambda_k(\ell) = \Lambda_k\, g(\ell)/g$ with Λ_k the sign-rule value. Axis k ignites at the slope $\ell_c^{(k)}$ where $\Lambda_k(\ell_c^{(k)}) = 1$, which exists only if $\Lambda_k > 1$. Raising ℓ at fixed β therefore moves the population along the same axis as raising β at fixed policy.

a. The mixed equilibrium is unstable under payoff learning. Let agent i use slope ℓ in context m, as its partners do. Its expected encounter payoff $\bar{U}_i$ changes with its own slope, at fixed partner policy and fixed covariance, as

$$\partial_\ell \bar{U}_i = \mathbb{E}_\Delta\big[\partial_\ell \bar{\pi}_\ell(\Delta)\,(3 - 6q)\big] = \tfrac{3}{2}\,\mathbb{E}[\Delta^2]\,\ell + O(\ell^3), \tag{S5}$$

with Δ the pairwise status difference, $q = \bar{\pi}_\ell(-\Delta)$ the partner's Go-probability of Sec. S1 C, $u_m = |\mathbf{g}_m|^2$, and $v_m = \hat{\mathbf{g}}_m^\top \mathbf{C} \hat{\mathbf{g}}_m$. The first form holds because the Go and Defer returns of Sec. S1 C differ by $3 - 6q$. The second expands both factors to first order in ℓ, $\partial_\ell \bar{\pi}_\ell \approx \Delta/2$ and $3 - 6q \approx 3\ell\Delta$. Where the scores are linear and pairing is random, $\mathbb{E}[\Delta^2] = 2u_m v_m$. The gradient vanishes at $\ell = 0$, so the mixed equilibrium is a stationary point. Its slope $\frac{3}{2}\mathbb{E}[\Delta^2]$ is positive whenever statuses differ, which the ω-identity noise ensures, so the stationary point is unstable and learning leaves it with no external bias. The first form, evaluated by quadrature over the read noise on simulated populations, agrees with $\frac{3}{2}\mathbb{E}[\Delta^2]\ell$ within 3.0% for $|\ell| \leq 0.12$, at $\beta = 2, 6, 12$ and in all three contexts (Fig. S1a). The linear-score value $3u_m v_m \ell$ exceeds it by up to 36% at $\beta = 12$, where the scores saturate.

b. Learning carries the population through the cascade. We simulated the dynamics of Sec. S10 with one slope ℓ_m per context, shared by all agents and updated at every step along the population average of each agent's REINFORCE estimate of Eq. (S5), from $\ell_m = 0.05$. A learned offset, $\tanh(\ell(s - \theta))$, stays within 0.09 of zero in every run and is omitted below. At $\beta = 6$ the slopes cross $\ell_c^{(1)} = 0.67$ at $t = 20$ and $\ell_c^{(2)} = 1.25$ at $t = 47$–50, in units of $1/\gamma$ and over two seeds. Axis 3 has $\Lambda_3 = 0.73$ and no threshold. At $\beta = 12$ all three axes cross in strength order, at $t = 5$–6, 10–13 and 60–66 (Fig. S1b). Each loop gain $\Lambda_m(\ell_m)$ rises toward its sign-rule value and stays below it, so learning crosses only the thresholds that β has placed within reach. Learning is slow compared with γ, and the covariance tracks its fixed-slope value: along the learning trajectory v_m deviates from it by a median 1.1% in contexts 1 and 2 (Fig. S1b, inset). The final payoff at $\beta = 6$ is 2.932 per encounter, against 2.969 under the sign rule. Below every threshold, at $\beta = 2$, the slopes still grow and the payoff reaches 2.554, against the Nash value 2.5.

c. The orientation of the convention is not selected. At fixed covariance the payoff is even in ℓ, because $\bar{\pi}_{-\ell}(\Delta) = \bar{\pi}_\ell(-\Delta)$ and the distribution of Δ is symmetric. The gradient of Eq. (S5) is then odd in ℓ and fixes no sign. The rule with $\ell < 0$, anti-follow, is the recommendation of Sec. S1 C reversed, and it is obeyed against a partner who also follows it, for the same reason. Its write-back is restoring: the higher-status agent Defers and is moved down, the lower-status agent plays Go and is moved up. At $\beta = 6$ and $\ell = -20$ this holds v_1 at 0.043, against 0.135 at $\ell = 0$. The signal stays weak, and anti-follow pays 2.539 per encounter against 2.969 at $\ell = +20$. It is nevertheless locally stable, since the myopic gradient still increases $|\ell|$. From $\ell = 0$ the orientation is decided by fluctuations: 11 of 30

context-runs on 10 seeds ended in follow, and 23 of 36 on 12 further seeds, together 34 of 66. All three runs started at $\ell = -0.5$ stayed in anti-follow. Selection between populations does not reverse it either. An evolution strategy on a scalar slope, ascending finite-difference estimates of the long-run payoff across replica populations of 500 agents, moved from $\ell = -0.5$ to -1.27 in 20 iterations. The long-run payoff rises with $|\ell|$ on both branches (to within 0.001, the Monte Carlo scale), so no local comparison favors follow. Selten showed that when both players see which role each holds, no mixed strategy is evolutionarily stable [8]. The Go/Defer game is a Hawk–Dove game with status as the role asymmetry. Follow and anti-follow are its Bourgeois and anti-Bourgeois conventions, which are both stable when the asymmetry does not enter the payoffs [9, 10].

d. A payoff-relevant asymmetry selects the orientation. Let a collision, (Go, Go), pay agent i the amount $-1 + c_{\mathrm{col}} \tanh(\Delta/\Delta_0)$, with Δ its status difference to the partner. The added term is antisymmetric within the pair, so the pair's total payoff, and with it W of Eq. (S2), is unchanged. The higher-status agent only loses less in a collision. The tilt adds $c_{\mathrm{col}} \tanh(\Delta/\Delta_0)\, q$ to the factor $3 - 6q$ of Eq. (S5). At $\ell = 0$, where $\partial_\ell \bar{\pi}_\ell = \Delta/2$ and $q = \frac{1}{2}$,

$$\partial_\ell \bar{U}_i\big|_{\ell=0} = \frac{c_{\mathrm{col}}}{4}\, \mathbb{E}_\Delta\big[\Delta \tanh(\Delta/\Delta_0)\big] > 0. \tag{S6}$$

The mixed equilibrium is no longer stationary, and learning leaves it toward follow. Balancing this term against the coordination term $\frac{3}{2}\mathbb{E}[\Delta^2]\,\ell$ for $|\Delta| \ll \Delta_0$ places the edge of the anti-follow basin at

$$\ell^* \approx -\frac{c_{\mathrm{col}}}{6\Delta_0}. \tag{S7}$$

Anti-follow remains locally stable for $\ell < \ell^*$. At $\Delta_0 = 0.5$ and $\beta = 6$ the measured edges in context 1 are $|\ell^*| = 0.069$, 0.141, 0.296, 0.699 for $c_{\mathrm{col}} = 0.25$, 0.5, 1, 2, against 0.083, 0.167, 0.333, 0.667 from Eq. (S7) (Fig. S1d). From $\ell = 0$, with $c_{\mathrm{col}} = 0.1$, 29 of 30 context-runs ended in follow, the exception in the weakest context. At every c_{col} from 0.25 to 1 all 30 did (Fig. S1c). On 10 further seeds at $c_{\mathrm{col}} = 0.1$, all 30 context-runs ended in follow. At $\Delta = \Delta_0$ a tilt of 0.1 changes the collision payoff by 0.076, against a collision cost of 1. In every run, axis 1 crossed its threshold before axis 2. An established anti-follow convention persists. From $\ell = -0.5$ learning reached follow only at $c_{\mathrm{col}} = 2$, where $|\ell^*| = 0.70 > 0.5$. One further seed each at $c_{\mathrm{col}} = 1$ and 2 reproduced both outcomes. From $\ell = -3$ it stayed in anti-follow at every c_{col} up to 2.

Role-following therefore emerges from the mixed equilibrium under payoff learning, provided status bears, however weakly, on the outcome of a collision. Local learning does not undo an established anti-follow convention unless $|\ell| < c_{\mathrm{col}}/6\Delta_0$. These results hold at fixed schemas (Model 4 of Table S1), random pairing, $T = 1$, and one slope per context shared by all agents. The convention runs are at $\beta = 6$. Agents that learn independently, and the incentive of the lower-paying role to climb in status, are not treated. Every number in this subsection is recomputed from the archived simulation data by `code/learn/numbers_learning.py`.

S2. PAIRING STATISTICS AND THE ENGINE PARAMETER

A. Beyond mean field: pairwise error rates

The mean-field error rate $P_{e,im} = \Phi(-|\tanh(\boldsymbol{\omega}_i \cdot \mathbf{g}_m)|/\sigma_{\mathrm{read}})$ replaces the partner's score by its zero mean. The true pairwise rate is $P_{e,ijm} = \Phi(-|\Delta_{im}(j)|/\sigma_{\mathrm{read}})$. This is exact at the bifurcation (all ω-identities near the origin) but breaks in the thick phase, where partners are drawn from a bimodal distribution and the status difference depends on whether the pair is same- or cross-mode. Resource accumulation therefore depends on the pairing distribution $\mathcal{P}_i$: $\dot{r}_i = \sum_m f_m \, \mathbb{E}_{j\sim\mathcal{P}_i}[W_{ijm}] - c\, r_i$. Three schemes span the range: uniform random pairing, which supplies a partner uncorrelated with one's own status; rank-ordered pairing, which supplies the status-mirrored (oracle-assisted) partner; and network-based pairing, which interpolates via the network's assortativity. Better pairing lowers the effective β_c and acts as a multiplier on $\bar{\pi}'(0)$: two societies with identical $\{\boldsymbol{\omega}_i\}$ but different pairing have different resource distributions, different $\mathbf{C}_r$, and potentially different repertoires. The pairing factor of each scheme, $(1 - \rho_{\mathrm{pair}})$, is derived below directly from this partner correlation, without reference to the thick-phase bimodal picture, so that it is equally well defined below threshold, where it sets β_c (Sec. S2 B).

B. Pairing efficiency as a partner correlation

The description of the pairing factor via "wasted same-mode encounters" is a supercritical intuition: it presupposes a bimodal population to have same- and cross-mode encounters at all. Since the factor sets β_c, the threshold below which no such modes exist, this description

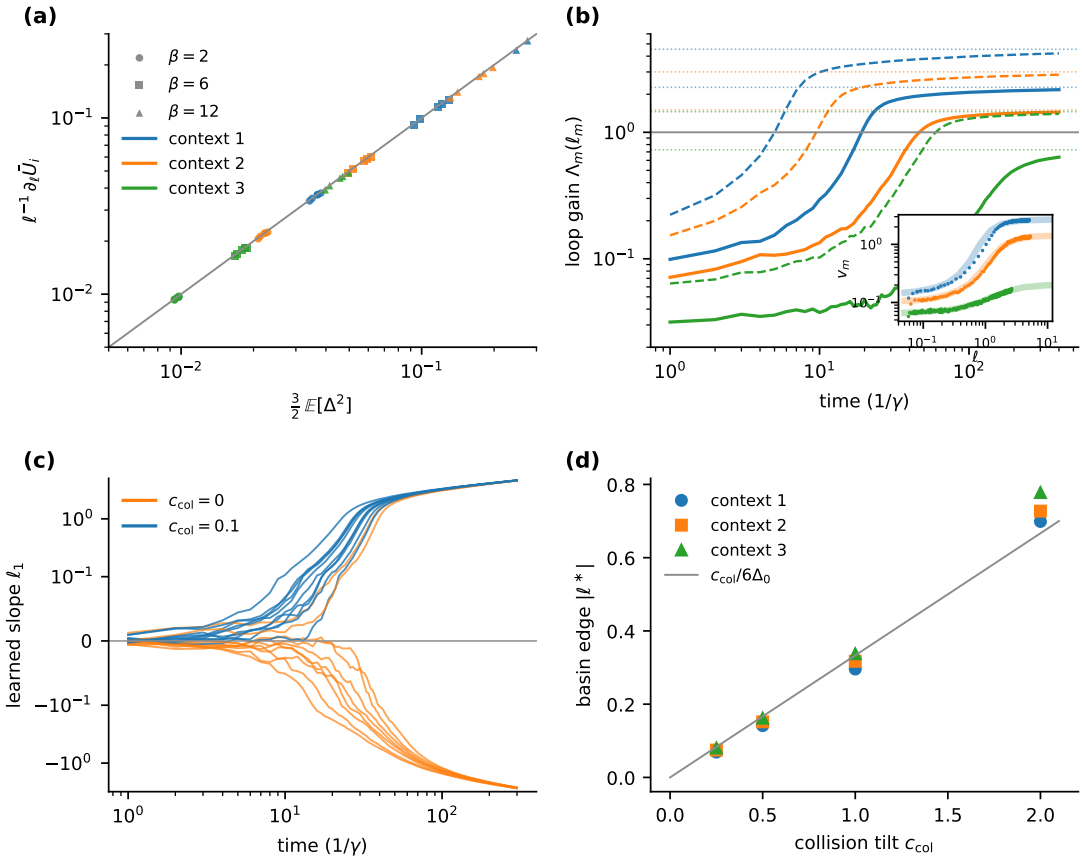


FIG. S1. **Learning the policy slope.** Dynamics of Sec. S10 with the policy family of Eq. (S4). **(a)** Myopic payoff gradient divided by ℓ, from quadrature over the read noise on simulated populations at fixed $|\ell| \leq 0.12$, against the small-slope prediction $\frac{3}{2}\mathbb{E}[\Delta^2]$ of Eq. (S5). Markers give β, colors the context, and the line is the identity. **(b)** Loop gain $\Lambda_m(\ell_m(t))$ of the learned slopes, from $\ell_m = 0.05$, at $\beta = 6$ (solid) and $\beta = 12$ (dashed). The gray line marks the threshold $\Lambda = 1$, and dotted lines the sign-rule values Λ_m. Inset: v_m against ℓ along the $\beta = 6$ trajectory (dots) and at fixed slope (wide lines). **(c)** $\ell_1(t)$ from $\ell = 0$ at $\beta = 6$, 10 seeds without the collision tilt (orange) and with $c_{\rm col} = 0.1$ (blue). **(d)** Edge of the anti-follow basin against $c_{\rm col}$, from the zero of the myopic gradient at fixed $\ell \in [-4, 0]$ ($\Delta_0 = 0.5$, $\beta = 6$). The line is Eq. (S7).

cannot be its defining one. We give the general definition here, and recover the bimodal intuition as its supercritical special case.

For a fixed context m, let $x_i = \boldsymbol{\omega}_i \cdot \mathbf{g}_m$ and, at a given encounter, let x_j be the projection of the partner the scheme supplies to i. Whatever the population's current dispersion—arbitrarily narrow and unimodal included—the pairing scheme induces some correlation

$\rho_{\text{pair}} = \text{Corr}(x_i, x_j) \in [-1, 1]$ between the two, taken over encounters at fixed time and context, with i drawn from the population and j the partner supplied. Since both are drawn from the same population they have the same marginal distribution, so $\text{Var}(x_j) = \text{Var}(x_i) = u_m v_m$ with $v_m = \hat{\mathbf{g}}_m^\top \mathbf{C} \hat{\mathbf{g}}_m$. The correlation is a property of the matching rule rather than of any pre-existing role structure. Treating (x_i, x_j) as jointly Gaussian with this correlation (exact near the origin, where the linearization Eq. (2) of the main text already applies), the partner's expectation conditional on the focal agent is $\mathbb{E}[x_j \mid x_i] = \rho_{\text{pair}} x_i$, so the expected difference is

$$\mathbb{E}[\Delta \mid x_i] = x_i - \mathbb{E}[x_j \mid x_i] = (1 - \rho_{\text{pair}})\, x_i. \tag{S8}$$

Since the linear response of the mean action is $g\, \mathbb{E}[\Delta \mid x_i]$, the pairing scheme enters the gain only through the factor $(1 - \rho_{\text{pair}})$, and the social loop gain of Eq. (5) of the main text and its thresholds are

$$\mathbf{\Lambda} \propto (1 - \rho_{\text{pair}}), \qquad \Lambda_k = \Lambda\, \mu_k, \qquad \text{ignition at } \Lambda_k = 1. \tag{S9}$$

Three cases recover the model ladder. *Random pairing* draws j independently of i, so $\rho_{\text{pair}} = 0$ exactly, at every population size, for factor 1 and $\beta_c = \gamma/\mu$, the Model 3 (mean-field) threshold. *Assortative (homophilic) pairing* supplies a partner resembling oneself, $\rho_{\text{pair}} \to 1$, factor 0. *Rank-ordered pairing*—pairing the agent at rank k with the agent at the mirrored rank $N_{\text{a}} + 1 - k$ along $\mathbf{g}_m$—induces $\rho_{\text{pair}} \to -1$ as $N_{\text{a}} \to \infty$. For a symmetric distribution the empirical quantile function is asymptotically antisymmetric, $x_{(N_{\text{a}}+1-k)} \to -x_{(k)}$, so the mirrored partner's projection converges to $-x_i$. This gives factor 2 and $\beta_c = \gamma/2\mu$, the Model 2 (oracle/reciprocal) threshold. A matcher need not represent roles to find this arrangement. Since W_{ijm} grows with $|\Delta_{im}(j)|$, the matching that maximizes realized payoff is the one that pairs each rank with its mirror, so a scheduler that optimizes on observed payoffs from past pairings alone arrives at $\rho_{\text{pair}} \to -1$ without any representation of status or role. We verified this numerically (Gaussian population, 10^2–10^4 realizations per N_{a}): the cross-sectional correlation of rank-ordered partners is $\rho_{\text{pair}} = -0.92$ at $N_{\text{a}} = 10$, -0.992 at $N_{\text{a}} = 200$, and -0.9996 at $N_{\text{a}} = 5000$, with the residual $1 - |\rho_{\text{pair}}|$ shrinking approximately as $N_{\text{a}}^{-0.9}$. Random pairing gives ρ_{pair} statistically indistinguishable from zero at every N_{a} tested. The conditional-expectation relation Eq. (S8) itself was confirmed by direct simulation of correlated Gaussian pairs across $\rho_{\text{pair}} \in \{-0.99, -0.5, 0, 0.5\}$, matching the predicted slope $1 - \rho_{\text{pair}}$ to three decimal places in every case.

Because ρ_{pair} is a property of the matching rule applied to whatever status distribution currently exists—not of a population already sorted into roles—Eq. (S9) is exactly as well defined arbitrarily far below threshold as above it. This closes the gap left by the thick-phase description. The "wastes same-mode encounters" language is recovered as the supercritical rereading of the same constant: relative to the rank-ordered ceiling (factor 2), an uncorrelated partner delivers half the difference, so its factor is 1. But that number is fixed already at $\rho_{\text{pair}} = 0$, independent of whether modes exist. Network-based pairing interpolates continuously between the extremes via whatever partner correlation the network's assortativity induces. Every ignition threshold, through Λ, therefore depends continuously on the choice of pairing scheme. This is the dependence the platform control of Sec. S2 D acts through.

a. The same correlation in the resource sector. The gain is not the only place ρ_{pair} enters. Near threshold the coordination payoff is quadratic in the status difference, $W_{ijm} \approx w_0 g^2 (x_i - x_j)^2$, so for the jointly Gaussian pair of Eq. (S8),

$$\mathbb{E}\big[(x_i - x_j)^2 \mid x_i\big] = (1 - \rho_{\text{pair}})^2 x_i^2 + (1 - \rho_{\text{pair}}^2)\, u_m v_m. \tag{S10}$$

The agent-specific and population-baseline terms of the resource income, Eq. (15) of the main text, therefore acquire *different* factors. Writing $\boldsymbol{A} = \sum_m f_m \mathbf{g}_m \mathbf{g}_m^\top$ and evaluating the fourth moment for Gaussian p^* by Wick's theorem,

$$\begin{gathered}\mathbb{E}[r^* \boldsymbol{\omega}\boldsymbol{\omega}^\top] \propto (1 - \rho_{\text{pair}})^2 \big[\operatorname{tr}(\mathbf{C}\boldsymbol{A})\, \mathbf{C} + 2\, \mathbf{C}\boldsymbol{A}\mathbf{C}\big] + (1 - \rho_{\text{pair}}^2) \operatorname{tr}(\mathbf{C}\boldsymbol{A})\, \mathbf{C}, \\ \mathbb{E}[r^*] \propto 2(1 - \rho_{\text{pair}}) \operatorname{tr}(\mathbf{C}\boldsymbol{A}),\end{gathered} \tag{S11}$$

and the ratio of the two leaves the single factor $(1 - \rho_{\text{pair}})$ of Eq. (17) of the main text, since $(1 - \rho_{\text{pair}})^2$ and $(1 - \rho_{\text{pair}}^2)$ sum to $2(1 - \rho_{\text{pair}})$. Two limits follow. Assortative matching ($\rho_{\text{pair}} \to 1$) returns $\mathbf{C}_r = \mathbf{C}$: the resource weighting disappears along with the coordination it rewards, and λ_m reduces to the unweighted variance v_m. The main text reaches the same collapse by setting $r_i \equiv 1$. Rank-ordered matching ($\rho_{\text{pair}} \to -1$) doubles the anisotropy. The matching rule therefore sets both the threshold and the strength of the selection that runs above it.

b. The role-free resource baseline. Averaging r^* over the population fixes the scale that the normalization of $\mathbf{C}_r$ divides out. For Gaussian p^* the agent-specific term of Eq. (S10) averages to $\mathbb{E}[(\boldsymbol{\omega} \cdot \mathbf{g}_m)^2] = \mathbf{g}_m^\top \mathbf{C} \mathbf{g}_m = u_m v_m$, so the two factors again sum to $2(1 - \rho_{\text{pair}})$ and

$$\bar{r}_{\text{rf}} \equiv \mathbb{E}[r^*] = \frac{2 w_0 g^2 (1 - \rho_{\text{pair}})}{c} \sum_m f_m\, u_m v_m = \frac{2 w_0 g^2 (1 - \rho_{\text{pair}})}{c} \operatorname{tr}(\mathbf{C}\boldsymbol{A}).$$

$\bar{r}_{\mathrm{rf}}$ is $O(\sigma_\omega^2)$ through $\mathbf{C}$, so agents accumulate resources from fluctuation status differences alone, before any role structure exists. It also appears as the denominator of $\mathbf{C}_r$. Since $\mathbf{\Lambda} = \Lambda \boldsymbol{A}$, the $\mathrm{tr}(\mathbf{C}\mathbf{\Lambda})$ of Eq. (17) of the main text is $\Lambda\,\mathrm{tr}(\mathbf{C}\boldsymbol{A})$, which is $\bar{r}_{\mathrm{rf}}$ up to the constants above, and with the play frequencies uniform and absorbed as in the main text the sum $\sum_n u_n v_n$ of Eq. (18) is $\mathrm{tr}(\mathbf{C}\boldsymbol{A})$ itself.

C. The three factors

The scalar critical condition $\beta > \gamma/g$ bundles cognition with noise, since the naive agent has $\bar{\pi}'(0) = 1/(\sigma_{\mathrm{read}}\sqrt{2\pi})$. Factoring into independently tunable contributions, define the agent inference efficiency

$$\eta_{\mathrm{agent}} \equiv \sigma_{\mathrm{read}} g/2, \tag{S12}$$

so that $\eta_{\mathrm{agent}} = 1/\sqrt{2\pi}$ (single sample) or $\sqrt{T/2\pi}$ (memory T), and take the pairing gain factor $(1-\rho_{\mathrm{pair}})$ from Sec. S2 B: 1 (random, the reference), 2 (rank-ordered, $N_{\mathrm{a}} \to \infty$), 0 (assortative). The effective gain $G_{\mathrm{eff}} = 2\eta_{\mathrm{agent}}(1-\rho_{\mathrm{pair}})/\sigma_{\mathrm{read}}$ gives the ignition condition

$$\Lambda \equiv \frac{2\beta}{\gamma} \cdot \frac{\eta_{\mathrm{agent}}\,(1-\rho_{\mathrm{pair}})}{\sigma_{\mathrm{read}}} > 1, \tag{S13}$$

separating ω-identity persistence β/γ (write/retention path), cognitive capacity η_{agent} (read path), and channel fidelity $(1-\rho_{\mathrm{pair}})/\sigma_{\mathrm{read}}$.

a. On the two appearances of σ_{read}. The observation noise appears twice in Eq. (S13)—inside $\eta_{\mathrm{agent}} = \sigma_{\mathrm{read}} g/2$ and in the denominator $(1-\rho_{\mathrm{pair}})/\sigma_{\mathrm{read}}$—but this is not double-counting. The physical noise enters exactly once, through $\bar{\pi}'(0) \propto 1/\sigma_{\mathrm{read}}$. Multiplying by σ_{read} in the definition of η_{agent} *removes* that dependence, which leaves a purely cognitive quantity ($1/\sqrt{2\pi}$ for the naive agent, independent of σ_{read}). The compensating $1/\sigma_{\mathrm{read}}$ in the information-supply factor restores it. The two factors are reciprocal and come from the same definition. They cancel, and $\Lambda \propto \beta\bar{\pi}'(0)/\gamma$ contains σ_{read} to the first power in the denominator, as it must. The grouping is chosen so that η_{agent} is a property of the agent alone (measurable from agent evaluations) and σ_{read} sits with $(1-\rho_{\mathrm{pair}})$ as a property of the channel, set by the communication layer.

b. Non-uniqueness. At linear order Λ is a loop gain (a Barkhausen criterion): the product of stage gains around the cycle (decision→write β; persistence $1/\gamma$; encounter selection $(1-\rho_{\mathrm{pair}})$; channel $1/\sigma_{\mathrm{read}}$; inference $\bar{\pi}'(0)$). Only the product is physical, so factors

regroup freely: a thermodynamic grouping isolates the dimensionful self-generated signal $\beta/(\gamma\sigma_{\text{read}})$ from a dimensionless per-cycle efficiency $\eta_{\text{agent}}(1-\rho_{\text{pair}})$ (the quantity bounded by the measurement-quality limit of Ref. [11]); alternatively $(1-\rho_{\text{pair}})$ absorbs into an effective channel $\sigma_{\text{read}}/(1-\rho_{\text{pair}})$, inference $\eta_{\text{agent}}(1-\rho_{\text{pair}})$, or coupling $\beta(1-\rho_{\text{pair}})$ (a bond-diluted mean-field ferromagnet, where J and occupation p enter T_c only through pJ). The degeneracy is a linear-order statement and breaks beyond threshold: β sets the thick-phase amplitude ($\gamma x(1+x^2)=\beta$ fixes x^* independently of ρ_{pair}) while ρ_{pair} sets how often separation converts to payoff. Societies at equal Λ but different $(\beta,\rho_{\text{pair}})$ therefore differ in order parameter, error rate, and hence in the resource flows entering $\mathbf{C}_r$. The grouping used here separates functionally distinct, independently measurable subsystems—the write path β/γ, the read path η_{agent}, and the channel $(1-\rho_{\text{pair}})/\sigma_{\text{read}}$. That separation makes the offline audit of Λ possible.

The measurement-quality bound of Ref. [11] constrains the read stage $\eta_{\text{agent}}(1-\rho_{\text{pair}})$, limiting the work extracted per measurement of an exogenous signal. Here the measured signal is not exogenous: role-play inscribed through β becomes the status that is read at the next encounter. The read-side ceiling of Ref. [11] therefore applies unchanged, and the write path is an additional stage that an engine with an external measured coordinate does not have. The two memories are distinct stages of the cycle: evidence integration over T samples, entering η_{agent}, and role persistence over $1/\gamma$, entering β/γ.

c. Substitutability and the cognitive floor. The factors substitute: good matching compensates for agents of low inference efficiency and vice versa, the social content of the measurement-quality bound. Substitution saturates, however—solving $\Lambda=1$ for the minimum inference gives, even at the rank-ordered maximum ($\rho_{\text{pair}}\to-1$, factor 2),

$$\eta_{\text{agent}}^{\min}=\frac{\sigma_{\text{read}}}{4(\beta/\gamma)}, \tag{S14}$$

an irreducible cognitive floor below which no orchestration sustains structure.

d. Vector case. For M contexts the scalar Λ becomes a matrix condition. Linearizing the mean-field drift (Appendix A of the main text), each context enters the gain with a weight equal to its play frequency f_m ($\sum_m f_m=1$, how often it is played) times its pairing factor $(1-\rho_m)\in[0,2]$, set by that context's partner correlation. These are the same diagonal reweighting of the schema second-moment matrix and therefore multiply,

$$\boldsymbol{\Lambda}=\frac{2\beta\eta_{\text{agent}}}{\gamma\sigma_{\text{read}}}\,\boldsymbol{H}^{\top}\operatorname{diag}(\mathbf{f})\,\boldsymbol{D}_{\rho}\,\boldsymbol{H},\qquad \boldsymbol{D}_{\rho}=\operatorname{diag}(1-\rho_1,\ldots,1-\rho_M)\equiv\operatorname{diag}(\boldsymbol{\phi}), \tag{S15}$$

writing $\phi_m \equiv 1 - \rho_m$ for the pairing efficiency of context m, the name used for this diagonal in the main text. The doubly-uniform case $f_m = 1/M$, $\rho_m = \rho_{\text{pair}}$ makes the diagonal proportional to the identity, and, with the frequency $1/M$ folded into β as in the main text, recovers the $\mathbf{\Lambda}$ of Eq. (5) of the main text. Axis k then activates when $\Lambda_k = (2\beta/\gamma)(\eta_{\text{agent}}(1-\rho_{\text{pair}})/\sigma_{\text{read}})\|g_k\|^2_{\text{eff}} > 1$, with $\|g_k\|^2_{\text{eff}}$ the k-th eigenvalue of $\boldsymbol{H}^\top\boldsymbol{H}$. The active-axis count is $M_{\text{eff}} = \#\{k : \Lambda_k > 1\}$, and the *ignition frontier* is the surface $\Lambda_k = 1$ in $(\beta/\gamma, \eta_{\text{agent}}, (1-\rho_{\text{pair}})/\sigma_{\text{read}})$ space.

D. Pairing as a control on the cascade class

The diagonal $\boldsymbol{D}_\rho = \text{diag}(1-\rho_m)$ is set by the matching rule, which fixes the partner correlation in each context and is controlled by the platform. The cascade class is fixed by the spectrum of $\mathbf{\Lambda} \propto \boldsymbol{H}^\top\boldsymbol{D}_\rho\boldsymbol{H}$ (Sec. III of the main text), so setting per-context pairing priorities reshapes that spectrum and hence the class. Uniform pairing ($\boldsymbol{D}_\rho = (1-\rho_{\text{pair}})\mathbf{I}_{M\times M}$) rescales $\boldsymbol{H}^\top\boldsymbol{H}$ and leaves the eigenvalue ratios unchanged, and with them the bifurcation spacings $\beta_c^{(k+1)}/\beta_c^{(k)}$; a non-uniform schedule does not. Down-weighting the contexts that load a coherent (spiked) direction transfers eigenvalue weight from the spike into the bulk, converting the signature of class (v), one onset then an avalanche, into the more even geometric sequence of class (ii).

This control carries a cost in coordination efficiency. Take the schedule $\{\rho_m\}$ as the control and the variance of the log-spacings $\ln(\mu_k/\mu_{k+1})$ as an inverse monitorability objective, zero variance being a perfectly geometric spectrum. A representative spiked repertoire ($M = d = 6$, schemas clustered about a common axis) is then optimized from log-spacing variance 1.35 under uniform pairing to 0.58, a spectrum a monitor can resolve. The cost is coordination: the schedule that achieves this reduces the aggregate pairing factor $\sum_m(1-\rho_m)$ by $\approx 40\%$, and since the coordination payoff scales with $(1-\rho_{\text{pair}})$ (Eq. (S2) and the resource flow), this is a direct loss of the engine's useful output. Monitorability is therefore obtained at the expense of coordination efficiency. The operator's choice is a point on that tradeoff rather than a free improvement. Because the reweighting acts through the same $\boldsymbol{H}^\top\boldsymbol{D}_\rho\boldsymbol{H}$ that sets ignition, an aggressive schedule also raises the effective $\beta_c^{(k)}$, coupling the control of the class weakly to the control of ignition.

S3. ROLE CAPACITY AND ASSOCIATIVE-MEMORY STRUCTURE

The vector feedback Eq. (2) of the main text is the negative gradient of the energy Eq. (12) of the main text, so single-agent dynamics are noisy gradient descent on a landscape whose stored patterns are the schemas $\{\mathbf{g}_m\}$ and whose activation is the integrated policy Ψ ($\Psi' = 2\bar{\pi} - 1$). This is the structure of a dense associative memory [12]. The feature/prototype duality is realized as two regimes: a *feature mode* (soft π, high σ_{read}, graded projections, near the bifurcation) and a *prototype mode* (sharp π, low σ_{read}, categorical ω-identities at role-polytope vertices, deep in the thick phase).

a. Where the mapping breaks. A social role is not an individual's alignment but a *bimodal split* of the population along $\mathbf{g}_m$, ordered by the variance $\mathbf{g}_m^\top \mathbf{C}\mathbf{g}_m$. Roles are relational. Three features absent from Hopfield retrieval follow: an anti-coordination constraint (the population must occupy complementary basins, not all retrieve one pattern); a self-generated signal (the "input" difference $\Delta_{im}(j)$ is produced by the population, so the landscape depth is state-dependent through the pairing distribution); and pairing dependence (basin depth that matters is the coordination payoff W_{ijm}, not the single-agent energy).

b. The bound. Despite this, the Hopfield structure caps the number of well-separated minima. Krotov–Hopfield give $M_{\max}^{\text{Hopfield}} \sim d^{n-1}$, with n the activation order (policy sharpness). The effective role capacity is lower, since each split must also satisfy the engine conditions:

$$M_{\text{eff}} = \#\Big\{k:\ \beta > \beta_c^{(k)},\ F_k > (\kappa/\alpha)|\mathbf{g}_k|^2\Big\}\ \le\ M_{\max}^{\text{Hopfield}} \sim d^{n-1}. \tag{S16}$$

On the fixed orthogonal axes of the main text this count is the eigendirection count of Eq. (20) of the main text, bounded by d. The Hopfield ceiling exceeds that bound because stored patterns need not be orthogonal, so it binds only where schemas may overlap. Both bounds tighten with cognition: sharper inference raises n (deeper, more numerous basins) and $\bar{\pi}'(0)$ (lower $\beta_c^{(k)}$, more axes bifurcating). Agents of low policy sharpness sustain few role dimensions. The ceiling rises exponentially in n, so modest capability gains (memoryless $\to$ short-memory) support qualitatively more role structure. The derivative $\partial M_{\text{eff}}/\partial n$ is the sensitivity of collective complexity to individual capability—a quantity safety evaluations could track.

c. No global identity. The label is a property of an axis, not of an agent. On an orthogonal repertoire the energy Eq. (12) of the main text is even in each projection $\boldsymbol{\omega} \cdot \hat{\mathbf{g}}_m$

separately, since Ψ is even and the confinement depends on $\boldsymbol{\omega}$ only through $|\boldsymbol{\omega}|$. The sign correlation therefore vanishes exactly, $\mathbb{E}[\mathrm{sign}(\boldsymbol{\omega}\cdot\hat{\mathbf{g}}_m)\mathrm{sign}(\boldsymbol{\omega}\cdot\hat{\mathbf{g}}_{m'})]=0$ for $m\neq m'$, at every β and every M. The harmonic confinement is a sum of squared projections, so the energy separates into one term per axis and the amplitudes are independent as well. An agent holds N independent bits and takes the higher-status role in about half the active contexts. A uniformly high-status ω-identity is one of the 2^N equally likely labels and carries no more weight than any other, so a global identity exists as a realization but not as a distinguished state. Overlap is the only source of correlation in the closed loop. The projections stop being independent coordinates once the repertoire is coherent. Expressing $|\boldsymbol{\omega}|^2$ in them inverts the Gram matrix $\hat{\boldsymbol{H}}\hat{\boldsymbol{H}}^\top$ of the unit schema axes, which at first order in the overlaps contributes a coupling $\gamma\,(\hat{\mathbf{g}}_m\cdot\hat{\mathbf{g}}_{m'})(\boldsymbol{\omega}\cdot\hat{\mathbf{g}}_m)(\boldsymbol{\omega}\cdot\hat{\mathbf{g}}_{m'})$ favoring equal signs where the overlap is positive. This is transposability read on the agent: a schema carrying across two contexts correlates the roles its holder takes in them. Below threshold the correlation is exact for any repertoire, since p^* is Gaussian with the covariance of Eq. (13) of the main text,

$$\mathbb{E}\big[\mathrm{sign}(\boldsymbol{\omega}\cdot\hat{\mathbf{g}}_m)\,\mathrm{sign}(\boldsymbol{\omega}\cdot\hat{\mathbf{g}}_{m'})\big]=\frac{2}{\pi}\arcsin\frac{\hat{\mathbf{g}}_m^\top\mathbf{C}\,\hat{\mathbf{g}}_{m'}}{[(\hat{\mathbf{g}}_m^\top\mathbf{C}\hat{\mathbf{g}}_m)(\hat{\mathbf{g}}_{m'}^\top\mathbf{C}\hat{\mathbf{g}}_{m'})]^{1/2}}.$$

The matching rule cannot generate the correlation, since it enters the gain as the diagonal $\boldsymbol{D}_\rho$ of Eq. (S15). A status ordering that ran across contexts would require instead the payoff asymmetry the mean field averages out (Sec. VI of the main text).

S4. WHERE THE ω-IDENTITY IS KEPT

Three storage architectures are consistent with the same encounter model, and they differ in which parameters an operator holds and in how much of the loop transfers with them (Table S2). *Self-storage* is the microfoundation adopted here (Appendix A of the main text): each agent keeps and presents its own ω-identity. A *status ledger* is a platform-maintained authoritative record of the same quantity. A *platform estimate* is neither: the operator reconstructs it from the encounter log. The three are distinguished throughout by these names.

Under self-storage none of the engine's state need be held by the platform: statuses live in the agents, and an operator's native controls are only the channel factors ρ_{pair} and σ_{read}. A status ledger, meaning a platform-maintained record of agent ω-identities, is then not

TABLE S2. Storage architectures for the ω-identity. Controls are those an operator holds natively, before any intervention. "Transfers" is how much of the closed loop moves with the state.

Architecture	Operator's controls	Transfers	Monitor
Self-storage	$\sigma_{\text{read}},\ \rho_{\text{pair}}$	nothing; the agents hold the write path	needs an estimate
Status ledger	$\sigma_{\text{read}},\ \rho_{\text{pair}},\ \beta,\ \gamma$	write and read paths	direct
Platform estimate	$\sigma_{\text{read}},\ \rho_{\text{pair}}$	the ω-sector up to $\sigma_\omega \boldsymbol{\eta}_i$; not Eqs. (3)–(4) of the main text	with estimation error

a primitive of the model but a designed intervention. It acts on several factors of Λ at once. It lowers σ_{read}, since clean reads replace noisy presentations. It enables high-quality (anti-correlated, low-ρ_{pair}) matching, because rank-ordered pairing requires global status knowledge, so pairing quality and storage architecture are not independent. It enforces honest presentation institutionally. And it transfers the write and expiry parameters β and γ from the agents to the operator. A ledger therefore gives the operator access to the engine's parameters. This is both why platforms build such records and where the corresponding failure mode lies.

The third architecture keeps the state with the agents but reconstructs it. Appendix B of the main text derives what an operator logging encounters recovers: the ω-sector, up to the unobserved noise $\sigma_\omega \boldsymbol{\eta}_i$, and no further. Eqs. (3) and (4) of the main text are driven by payoff rather than by action, so a platform estimate yields the monitor at an estimation error while the reward path stays with the agents.

The controllable-coordination principle requires the operator retain the ability to modulate ρ_{pair} (including to the no-information value $\rho_{\text{pair}} = 1$). The same logic applies to memory. If ω-identity migrates out of the ledger—back to self-storage (an agent maintaining its own persona notes sets its own β) or distributed peer memory (statuses held only as other agents' beliefs)—control of β/γ passes from operator to collective, and role differentiation becomes correspondingly harder to reverse. The locus of ω-identity storage is therefore itself a monitorable, safety-relevant variable, complementary to the covariance-spectrum diagnostics: an early sign that the operator is losing those controls is not a change in any factor of Λ but a change in who sets them.

A. Repertoire seeding and the ledger's memory

Two accounting choices sit under the regime map: the dispersion of the seeded repertoire, which sets its horizontal axis, and the ledger's memory horizon, which does not. We treat them in turn, since the second is easily mistaken for the first.

a. What sets the horizontal axis. The ratio $r = r_{\mathrm{act}}/r_{\mathrm{comp}}$ is fixed by how unequally the candidate schemas are weighted when they are seeded (Eq. (26) of the main text): a repertoire launched at near-equal weights crosses threshold almost together (large r, churn, accelerating cascades), while one whose schemas are tiered by orders of magnitude crosses in a well-separated sequence (small r, entrenchment, decelerating cascades). For a deployed multi-agent system the operator defines the task taxonomy—which contexts exist, and how much traffic each is launched with. So σ_u is platform-owned in the same sense as the per-context frequencies f_m and pairing efficiencies ϕ_m of Sec. S2, and acts on the same object: the spectrum of the effective gain. This is also the point at which the open-repertoire process left outside the closed loop in the main text re-enters as a design variable rather than an exogenous accident.

b. What the ledger's memory does instead. Where a platform maintains the ledger, the resource decay rate c is likewise an accounting choice, but it does *not* move the map. The resource dynamics $\dot{r}_i = W_i(t) - c\, r_i$ (with $W_i = \sum_m f_m\, \mathbb{E}_j[W_{ijm}]$ the instantaneous coordination income) integrate to

$$r_i(t) = \int_{-\infty}^{t} e^{-c(t-t')}\, W_i(t')\, dt', \tag{S17}$$

so wealth is past income credited through an exponential memory kernel $K(\tau) = e^{-c\tau}$. The mean lifetime of a credited unit is $\int_0^\infty \tau K\, d\tau / \int_0^\infty K\, d\tau = 1/c$, i.e. $c = 1/\tau_{\mathrm{mem}}$ is the inverse memory horizon of the ledger: how long a past contribution keeps counting toward present standing. The exponential kernel is the unique memoryless (Markovian) choice consistent with the local ODE. A platform may adopt a different retention rule: a hard credit window of width T (mean lifetime $T/2$, so $c_{\mathrm{eff}} = 2/T$), or a power-law reputation decay $K(\tau) \sim \tau^{-\alpha}$ ($\alpha > 2$, mean lifetime $(\alpha-1)/(\alpha-2)$). Each realizes the same accumulation with c the effective inverse mean-lifetime of that kernel. Either way c is set by the accounting rule, not by physics.

The memory horizon $\tau_{\mathrm{mem}} = 1/c$ nonetheless leaves r alone, because $\mathbf{C}_r$ is normalized by $\sum_i r_i$. The stationary weights $r_i^* = c^{-1} \sum_m f_m W_{im}$ contain c as an overall factor, which

the normalization divides out, so neither the compounding rate r_{comp} nor the activation rate r_{act} inherits it. Lengthening the ledger's memory therefore does not by itself make the cascade more monitorable. It controls whose past counts instead: a long kernel credits contributions made long ago, so $\mathbf{C}_r$ is weighted by cumulative rather than current standing. The kernel length is therefore the horizon over which early advantage compounds into locked-in incumbency. The legibility/lock-in tension of the main text survives, but legibility and lock-in both act through the repertoire: where the budget caps growth, a widely dispersed seeding yields the extrapolable classes, and it also gives the earliest schemas the longest time to compound.

S5. MONOTONICITY OF THE PINNING FIXED POINT FOR GENERAL REPERTOIRES

The pinning argument of the closed-pool construction (Sec. IV A of the main text) requires that each schema's $\lambda_m = \hat{\mathbf{g}}_m^\top \mathbf{C}_r \hat{\mathbf{g}}_m$ increase with its strength $u_m = |\mathbf{g}_m|^2$, so that the shared $\lambda_m = \kappa/\alpha$ at an interior fixed point forces all survivors to equal strength. We verify this monotonicity here, first for an orthogonal repertoire and then in general.

a. Orthogonal repertoire. Here $\hat{\mathbf{g}}_m$ is an eigenvector of $\mathbf{C}_r$ and λ_m reduces to $v_m(1 + (1-\rho_{\mathrm{pair}})u_m v_m / \sum_n u_n v_n)$ with single-axis variance $v_m = \sigma_\omega^2/[2\gamma(1-\Lambda u_m)]$. Both factors increase with u_m:

$$\frac{\partial v_m}{\partial u_m} = \frac{\Lambda \sigma_\omega^2/2\gamma}{(1-\Lambda u_m)^2} > 0 \tag{S18}$$

throughout the sub-critical range $\Lambda u_m < 1$ (a stronger schema differentiates its adherents more, widening the status distribution along its axis), and the amplification factor $1 + (1-\rho_{\mathrm{pair}})u_m v_m / \sum_n u_n v_n$ likewise grows with u_m, so $\partial\lambda_m/\partial u_m > 0$.

b. Non-orthogonal repertoire. Here $\hat{\mathbf{g}}_m$ is no longer an eigenvector of $\mathbf{C}_r$ and λ_m is a Rayleigh quotient mixing several eigenvalues. Differentiating $\mathbf{C} = \frac{\sigma_\omega^2}{2\gamma}(\mathbf{I}_{d\times d} - \mathbf{\Lambda})^{-1}$ gives

$$\frac{\partial \mathbf{C}}{\partial u_m} = \frac{2\gamma\Lambda}{\sigma_\omega^2} (\mathbf{C}\hat{\mathbf{g}}_m)(\mathbf{C}\hat{\mathbf{g}}_m)^\top, \tag{S19}$$

a rank-one positive-semidefinite matrix for any repertoire geometry, so the leading contribution $\hat{\mathbf{g}}_m^\top \mathbf{C} \hat{\mathbf{g}}_m$ to λ_m rises with u_m unconditionally. The remaining contribution $(1-\rho_{\mathrm{pair}})\hat{\mathbf{g}}_m^\top \mathbf{C}\mathbf{\Lambda}\mathbf{C}\hat{\mathbf{g}}_m / \operatorname{tr}(\mathbf{C}\mathbf{\Lambda})$ contains a competing $1/\operatorname{tr}(\mathbf{C}\mathbf{\Lambda})$ that grows with u_m. Numerically the sum remains increasing across random, overcomplete ($M > d$), and strongly coherent

repertoires throughout the sub-critical range. The degenerate fixed point is therefore a saddle in the general case as well, and pinning converts the competitive-exclusion equal-fitness result into equal strength.

S6. VALIDITY OF THE LINEARIZATION AND THE FLUCTUATION-SHIFTED THRESHOLD

The subcritical stationary state is not the point $\boldsymbol{\omega} = 0$ but a finite-width Gaussian ensemble in which the state $\boldsymbol{\omega}_i$ has covariance $\mathbf{C} = \frac{\sigma_\omega^2}{2\gamma}(\mathbf{I}_{d\times d} - \boldsymbol{\Lambda})^{-1}$. Expanding "about $\boldsymbol{\omega} = 0$" is therefore justified not by concentration of the state but by two separate facts, which we set out here.

a. Stability is a Jacobian property. Ignition is the loss of linear stability of the symmetric fixed point $\boldsymbol{\omega} = 0$ of the deterministic mean-field drift $\mathbf{F}(\boldsymbol{\omega})$. Stability is governed by the Jacobian $\partial\mathbf{F}/\partial\boldsymbol{\omega}|_0$, a derivative evaluated *at* the fixed point. It makes no reference to where the probability mass of the noisy state sits. The eigenvalues of $\boldsymbol{\Lambda} - \mathbf{I}_{d\times d}$ change sign at $\beta_c^{(k)} = \gamma/[(1-\rho_{\text{pair}})\mu_k]$ regardless of how dynamical noise dresses the state, so the thresholds are the exact stability boundary of the deterministic flow.

b. Where the state's width enters. The single place the finite width matters is the partner. The focal response $\partial\langle a\rangle_m/\partial x_i$ averages over the partner projection $x_j = \mathbf{g}_m \cdot \boldsymbol{\omega}_j$. Writing $X = -x_j + \sigma_{\text{read}}\xi$ for everything random given x_i, the effective gain is $2f_X(0)$ with $\text{Var}(X) = v + \sigma_{\text{read}}^2$, $v \equiv \text{Var}(x_j) = \mathbf{C}_{mm}$. Gaussian X then gives an effective inference slope $1/[\sqrt{2\pi}\,(v+\sigma_{\text{read}}^2)^{1/2}]$ and a broadened single-mode gain

$$\mu_{\text{eff}}(v) = \frac{\mu}{\sqrt{1 + v/\sigma_{\text{read}}^2}}, \qquad \mu = g, \tag{S20}$$

recovering the bare μ only as $v \to 0$. This is the same finite partner variance that makes the mean-substitution $x_j \to \mathbb{E}[x_j \mid x_i]$ inside the nonlinear policy inexact away from onset (Sec. S2 B). Both corrections are controlled by the same v.

c. Self-consistency and the weak-noise limit. The partner variance is itself the mode variance, fixed self-consistently by the Ornstein–Uhlenbeck balance,

$$v = \mathbf{C}_{mm} = \frac{\sigma_\omega^2}{2\big(\gamma - \beta(1-\rho_{\text{pair}})\mu_{\text{eff}}(v)\big)} = O(\sigma_\omega^2) \;\text{ subcritically.} \tag{S21}$$

Because $v = O(\sigma_\omega^2)$ while β_c is independent of σ_ω, the cloud shrinks to a delta at the origin as $\sigma_\omega \to 0$ and $\mu_{\text{eff}} \to \mu$. The bare threshold $\beta_c = \gamma/[(1-\rho_{\text{pair}})\mu]$ is therefore exact in the weak-noise limit, where "expand about the mean" and "expand about the origin" coincide.

d. The fluctuation-shifted threshold. At finite σ_ω, expanding Eq. (S20) to leading order, $\mu_{\text{eff}} \approx \mu(1 - v/2\sigma_{\text{read}}^2)$, softens the gain and pushes onset up:

$$\beta_c(\sigma_\omega) = \frac{\gamma}{(1-\rho_{\text{pair}})\mu}\Big(1 + \frac{v}{2\sigma_{\text{read}}^2} + O(\sigma_\omega^4)\Big), \qquad v \approx \frac{\sigma_\omega^2}{2(\gamma - \beta(1-\rho_{\text{pair}})\mu)}. \tag{S22}$$

The correction is the mean-field analogue of a Ginzburg shift. It is $O(\sigma_\omega^2)$ and negligible except in a critical window where v grows to $O(\sigma_{\text{read}}^2)$, i.e. $|\gamma - \beta(1-\rho_{\text{pair}})\mu| \lesssim \sigma_\omega^2/\sigma_{\text{read}}^2$, whose width vanishes as $\sigma_\omega \to 0$. Within that window the divergence of the bare theory is rounded by the self-consistent feedback of Eq. (S21). The main text reports the $\sigma_\omega \to 0$ value, and the finite-noise shift and rounding are among the quantities a full-loop simulation could measure.

e. The invasion threshold and its fluctuation shift. A schema becomes active where its replicator rate, Eq. (4) of the main text, turns positive, $\alpha\lambda_m > \kappa + \varphi$. The correction term of Eq. (18) of the main text is nonnegative for $\rho_{\text{pair}} \le 1$, so $\lambda_m \ge v_m$, and both grow with u_m. The invasion strength $u_m^\dagger$ therefore satisfies

$$\Lambda u_m^\dagger \;\le\; 1 - \frac{\alpha\sigma_\omega^2}{2\gamma(\kappa+\varphi)} \;<\; 1,$$

with equality as the invading schema's share of $\sum_n u_n v_n$ vanishes. Since λ_m increases with u_m, $u_m^\dagger$ is unstable. Budget pressure raises it, so schemas already drawing on the channel lift the threshold a later entrant must exceed. The two thresholds coincide only as $\sigma_\omega \to 0$. Below every bifurcation the covariance sits at the noise floor of Eq. (13) of the main text rather than at zero, so $\lambda_m > 0$ and the strengths grow before any role structure exists. A positive $u_m^\dagger$ requires $\alpha\sigma_\omega^2/2\gamma < \kappa + \varphi$, and where that fails every seeded schema invades, so the sub-threshold state in which every schema decays holds only while the decay rate exceeds the fluctuation reinforcement.

f. No intrinsic ceiling. The bare reinforcement supplies no turnover of its own. Along a single active axis, well above threshold, the cross-role read is error-free, so the two roles sit at $\boldsymbol{\omega} \cdot \hat{\mathbf{g}} = \pm\beta\sqrt{u}/2\gamma$ ("Why the score saturates" below). The thick-phase strength is then $\lambda(u) \approx \beta^2 u/4\gamma^2$ plus the within-role variance. It grows linearly in u without turning over, so no *intrinsic* strength exists at which reinforcement saturates. The saturating score

caps what each agent presents, and the bounded action keeps $|\boldsymbol{\omega}_i|$ finite at fixed strength. Neither caps the schema *strength*, which keeps rising as the two roles separate. Both ceilings of the regime map are therefore imposed rather than emergent.

g. Why the score saturates. Sec. I of the main text places the loop's nonlinearity in the score $s(x)$, with $x = \boldsymbol{\omega} \cdot \mathbf{g}_m$. A linear score leaves the two-role state unstable above $\Lambda = 2$. Take one axis of an orthogonal repertoire under random pairing, with two roles at $x = \pm x^*$ well separated compared with σ_{read}. A partner from the other role is then read without error, and its read, ± 1, does not depend on either ω-identity. A partner from the agent's own role presents the status difference $s(x_i) - s(x_j) \approx s'(x^*)(\delta_i - \delta_j)$, with δ the deviation from x^*, and is read at slope g. Half of an agent's partners hold each role. The cross-role half sets the role position, $x^* = \beta u/2\gamma = \Lambda/2g$. The same-role half drives the deviation with the linear dynamics of the main text, at gain

$$\Lambda_{\text{w}} = \tfrac{1}{2}\,\Lambda\, s'(x^*).$$

For a linear score $\Lambda_{\text{w}} = \Lambda/2$. Above $\Lambda = 2$ the spread within each role grows, each role splits again, and the population spreads toward a uniform ranking. For $s = \tanh$, $\Lambda_{\text{w}} = \frac{1}{2}\Lambda \operatorname{sech}^2(\Lambda/2g)$, which never exceeds $0.45\,g$. The two roles are therefore stable at every Λ when $g < 2.2$. At $\sigma_{\text{read}} = 0.6$, $g = 1.33$ and $\Lambda_{\text{w}} \leq 0.60$. A simulation of one axis ($N_{\text{a}} = 2000$, 64 partners per step with the read noise averaged, harmonic restoring force) confirms both cases. The bimodality coefficient of x is 1 for two point masses and 5/9 for a uniform law. Under the linear score it peaks at 0.64 at $\Lambda = 2$ and falls to 0.57 by $\Lambda = 6$. Under the tanh score it rises to 0.99 (`code/sim_role_splitting.py`).

S7. SCOPE OF THE REGIME MAP

These qualify the regime map of Sec. IV B of the main text.

a. Near-orthogonal repertoires. The counting statements hold in that limit: orthogonal axes make $\mu_k = u_k$, so eigenvalues and strengths coincide and M_{cap} counts saturated schemas directly. Overlap breaks the identification. Schemas then couple through $\mathbf{C}_r$ as well as through the shared budget, which can be reached before the per-schema ceiling and leave survivors equalized below u_{cap}, so the boundaries degrade into crossovers as the repertoire becomes coherent. The width of that window is not fixed here. For random directions the

pairwise overlap is $O(d^{-1/2})$, and the identification survives while the accumulated shift stays below the strength gaps, $M/d \ll \Delta u/u$. Near-orthogonality is therefore the small-aspect-ratio corner of the group $q = M/d$ already held fixed in Sec. IV B of the main text. At $q = O(1)$ random directions give the Marchenko–Pastur spectrum of class (iv) instead, and at $q \gg 1$ they give a degenerate one, which no longer depends on the strength distribution at all.

b. Two classes are not endogenous fixed points. The Marchenko–Pastur bulk (iv) is the signature of a repertoire not yet organized by resource feedback (the natural initial condition for freshly seeded schemas, and transient under the loop), and its persistence in data would itself indicate the loop has not closed. The exact degeneracy of the saturated-cluster regime holds only within the Gaussian pinning argument. Generic cross-axis resource coupling and residual coherence broaden it into a narrow cluster.

c. Parametrizing the map. Since only the dimensionless ratio $\Theta = U_{\rm cap}/(M u_{\rm cap})$ enters the morphology, we parametrize the regime map by this ratio alone, rendered in Fig. 4 of the main text as the capacity $M_{\rm cap}/M = \Theta$, and never fix an absolute $u_{\rm cap}$. The only absolute scale retained is the seed-to-budget ratio $u_0/U_{\rm cap}$, which sets the log dynamic range $L = \ln(u_{\rm cap}/u_0)$ entering the saturated-cluster/geometric boundary. This is the seed *scale*, independent of the seed *dispersion* σ_u that fixes the horizontal axis. The remaining groups ($\alpha\langle\lambda\rangle/\kappa$, supercriticality Λ_1, aspect ratio M/d) are held fixed, as is the repertoire geometry $\{\hat{\mathbf{g}}_m \cdot \hat{\mathbf{g}}_n\}$, which the map takes near-orthogonal.

d. Sustaining a hierarchy. Holding one indefinitely requires continual injection of new schemas, an open-repertoire process outside the closed loop that we do not treat. Within the closed loop, the endogenous source of novelty during the u-transient is transposability, the schema overlap $\mathbf{g}_m \cdot \mathbf{g}_n$ through which the developing $\mathbf{C}_r$ raises successive axes to threshold.

S8. FROM THE ENCOUNTER PROCESS TO THE MEAN-FIELD DRIFT

Eqs. (1) and (2) of the main text are separated by two limits, $\nu \to \infty$ and $N_{\rm a} \to \infty$, which do different work. The average over the pairing probabilities belongs entirely to the first. The second is needed only to close the resulting equation.

a. The window average. Fix a window Δt with $1/\nu \ll \Delta t \ll 1/\gamma$. It holds $N_{\rm enc} = \nu\,\Delta t \gg 1$ encounters, over which the ω-identity moves by a fraction $O(\gamma\,\Delta t) \ll 1$ of its own

scale, so the encounters are draws from a fixed distribution. Each draws a context m with probability f_m, a partner j with probability $p_{ij}^{(m)}$, and T observation noises, independently of the others. The kick term of Eq. (1) of the main text is then $N_{\rm enc}$ times an empirical mean of the i.i.d. vectors $X_e = a_e\,\mathbf{g}_{m_e}$,

$$b_0 \sum_e a_e\,\mathbf{g}_{m_e} = b_0 N_{\rm enc}\Big(\frac{1}{N_{\rm enc}}\sum_e X_e\Big) = \beta\,\Delta t\,\big(\mathbb{E}[X] + O(N_{\rm enc}^{-1/2})\big), \tag{S23}$$

with $\beta = \nu b_0$. The X_e are bounded by $\max_m |\mathbf{g}_m|$, so the fluctuation is the $O(N_{\rm enc}^{-1/2})$ of the central limit theorem. It is the shot noise of Appendix A of the main text.

b. The noise average of a single decision. Fix the context m and the partner j, so the status difference $\Delta = \Delta_{im}(j)$ is a number. The agent draws T samples $s_t = \Delta + \sigma_{\rm read}\xi_t$ with ξ_t independent standard Gaussians, and their mean is Gaussian about the true difference,

$$\bar{s}_T = \frac{1}{T}\sum_{t=1}^{T} s_t \;\sim\; \mathcal{N}\Big(\Delta,\; \frac{\sigma_{\rm read}^2}{T}\Big). \tag{S24}$$

The action takes only two values, so its average over the observation noise is fixed by a single probability,

$$\bar{a}(\Delta) \equiv \mathbb{E}_\xi[a] = (+1)\,\Pr(a{=}{+}1 \mid \Delta) + (-1)\,\Pr(a{=}{-}1 \mid \Delta) = 2\bar{\pi}(\Delta) - 1, \tag{S25}$$

with $\bar{\pi}(\Delta) = \Pr(a{=}{+}1 \mid \Delta)$ the Go-probability of the policy at true difference Δ. Equation (S25) is an identity for any policy, deterministic or not. Only $\bar{\pi}$ carries the policy. For the rule $a = \mathrm{sign}(\bar{s}_T)$, Eq. (S24) gives $\bar{\pi}(\Delta) = \Phi(\sqrt{T}\,\Delta/\sigma_{\rm read})$ and so $\bar{a}(\Delta) = \mathrm{erf}\big(\sqrt{T}\,\Delta/(\sqrt{2}\,\sigma_{\rm read})\big)$, the form used in Appendix A of the main text. The T samples enter only through the variance of $\bar{s}_T$, which is why the response sharpens as $\sqrt{T}$. Policy symmetry gives $\bar{\pi}(-\Delta) = 1 - \bar{\pi}(\Delta)$, so $\bar{a}$ is odd, and the drift of Sec. II of the main text inherits that oddness.

c. Evaluating the expectation. Conditioning outward from the observation noise, the expectation factorizes,

$$\mathbb{E}[X] = \sum_m f_m\,\mathbf{g}_m \sum_j p_{ij}^{(m)}\,\bar{a}\big(\Delta_{im}(j)\big), \qquad \bar{a}(\Delta) \equiv 2\bar{\pi}(\Delta) - 1. \tag{S26}$$

The innermost average is Eq. (S25), taken at fixed context and partner. The partner and context averages follow. One can instead group the window's encounters by partner, in which case the count $N_{\rm enc} f_m p_{ij}^{(m)}$ of encounters with j in context m supplies the weight.

That route requires every cell to be well populated, $N_{\mathrm{enc}} f_m p_{ij}^{(m)} \gg 1$, which under uniform matching means $\nu/\gamma \gg N_{\mathrm{a}}$. The route through Eq. (S26) requires only $N_{\mathrm{enc}} \gg 1$ and gives the same weights.

d. The dynamics at finite N_{a}. Substituting Eq. (S26) into Eq. (S23) gives the dense-encounter limit at fixed population size,

$$\dot{\boldsymbol{\omega}}_i = -\gamma\, \boldsymbol{\omega}_i + \beta \sum_m f_m\, \mathbf{g}_m \sum_j p_{ij}^{(m)}\, \bar{a}\big(\Delta_{im}(j)\big) + \sigma_\omega \boldsymbol{\eta}_i. \tag{S27}$$

No randomness remains over which partners were drawn. The pairing probabilities already enter as weights, exactly as they do in Eq. (2) of the main text.

e. What remains at finite N_{a}. Equation (S27) is not a closed equation for $\boldsymbol{\omega}_i$. Its right-hand side carries the other $N_{\mathrm{a}} - 1$ ω-identities, each solving the same equation, and two obstructions survive.

First, $\boldsymbol{\omega}_j$ is not independent of $\boldsymbol{\omega}_i$. The two have met, and in an anti-coordination game their kicks are anti-correlated—when i goes, j defers—so $\boldsymbol{\omega}_j$ retains an imprint of the focal agent's own history. Encounters between the pair occur at rate $\nu p_{ij}^{(m)}$, and since ν is a fixed per-agent budget with $\sum_j p_{ij}^{(m)} = 1$, that rate is $O(\nu/N_{\mathrm{a}})$. The imprint is therefore $O(1/N_{\mathrm{a}})$ relative to $\boldsymbol{\omega}_j$'s own drive. An interaction budget growing with N_{a} would not have this property.

Second, the sum runs over the realized ω-identities, each a solution of a noisy equation. Since $|\bar{a}| \leq 1$, the sum has standard deviation $O(N_{\mathrm{eff}}^{-1/2})$ about its population value, where

$$N_{\mathrm{eff}} = \Big(\sum_j \big(p_{ij}^{(m)}\big)^2\Big)^{-1} \tag{S28}$$

is the participation ratio of the pairing kernel, equal to $N_{\mathrm{a}} - 1$ under uniform matching. The drift therefore fluctuates between agents and between runs.

f. The limit. The mean-field limit supplies $N_{\mathrm{eff}} \to \infty$, where both obstructions vanish: the empirical distribution of the other ω-identities converges to a fixed law, the partner decouples from the focal agent's history, and the sum over j becomes an expectation against that law. Where the matching rule correlates the partner's projection with the focal agent's, the relevant law is the conditional one, and near the symmetric fixed point it contributes the factor $(1 - \rho_{\mathrm{pair}})$ of Sec. S2 B. Writing that expectation as $\langle a \rangle_m(\boldsymbol{\omega}_i)$ gives Eq. (2) of the main text.

g. Averaging the difference and averaging the action. What the limit delivers is $\mathbb{E}_j[\bar{a}(\Delta_{im}(j))]$, an average of the *action* over the partner. Eq. (2) of the main text instead evaluates $\bar{a}$ at the averaged *status difference*. The two operations are not the same, and only the second carries an error.

Averaging the status difference is exact. $\Delta_{im}(j) = \tanh x_i - \tanh x_j$ is linear in the partner's score, so the average is $\tanh x_i - \mathbb{E}[\tanh x_j \mid x_i]$ with no approximation. Under random matching the conditional mean is the population mean score, which vanishes for a symmetric population. Near the symmetric fixed point the scores are linear, and the joint Gaussianity of (x_i, x_j) supplies $\mathbb{E}[x_j \mid x_i] = \rho_{\text{pair}} x_i$, hence the expected difference $(1-\rho_{\text{pair}})x_i$ of Sec. S2 B.

Averaging the action is not the same, because $\bar{a}$ is nonlinear. For this policy the exact partner average is available in closed form, through the same ω-identity that produced the erf in the first place: for z Gaussian with variance s^2,

$$\mathbb{E}_z\big[\operatorname{erf}(\kappa(u+z))\big] = \operatorname{erf}\Big(\frac{\kappa\, u}{\sqrt{1+2\kappa^2 s^2}}\Big). \tag{S29}$$

Smoothing an erf by a Gaussian returns an erf, with the two variances added in quadrature inside the argument. Applied to the observation noise it gives the noise average itself: the hard sign is the $\kappa \to \infty$ limit of $\operatorname{erf}(\kappa\,\cdot)$, and Eq. (S29) then returns $\bar{a}(\Delta) = \operatorname{erf}\big(\sqrt{T}\Delta/(\sqrt{2}\,\sigma_{\text{read}})\big)$, which is Eq. (S25) and Appendix A of the main text. Applied a second time, now to the partner projection, it softens the slope at the origin by $(1+2\kappa^2 v)^{-1/2}$ with v the partner variance. For a single evidence sample that factor is the $\mu_{\text{eff}}(v)$ of Eq. (S20).

The substitution made in Eq. (2) of the main text is the $s \to 0$ case of Eq. (S29). It is controlled where the partner variance is small against σ^2_{read}, and $v = O(\sigma_\omega^2)$ subcritically by Eq. (S21), so the substitution is exact as $\sigma_\omega \to 0$ and costs an $O(\sigma_\omega^2)$ shift of onset at finite noise (Eq. (S22)). Above threshold the partner distribution on that axis is bimodal, and neither the Gaussian conditional mean nor the small-v expansion is controlled there.

h. Why only one average is exchanged. The same $s \to 0$ limit taken on the observation noise would send $\sigma_{\text{read}} \to 0$, returning the hard sign with a delta-function slope and no finite gain to set a threshold. The two averages are therefore not on the same footing. The observation noise is smoothed and kept, and that smoothing is the model's read stage rather than an approximation. The partner spread is smoothed and then discarded. What

separates them is scale: σ_{read} is a parameter of the model, while $v = O(\sigma_\omega^2)$ follows from weak ω-identity noise. A second asymmetry is that ξ has a fixed distribution, known before the population is solved for, whereas the partner distribution is the unknown. Averaging over the latter exactly would leave the drift a functional of the population distribution rather than a closed equation in $\boldsymbol{\omega}_i$.

Note on scope. The derivations of Secs. II B and IV A of the main text are carried out in the Gaussian, near-threshold, mean-field regime, where the Wick reduction closes. That regime is the noise-dominated one: the ω-identity noise, not the drift, sets the dispersion along an axis below its threshold, and the population stays unimodal there. An axis above threshold is instead held at $\pm m$ by the drift and splits, so the closure fails axis by axis as the cascade proceeds. Their extension into the thick phase (bimodal partner distributions, saturated policies and scores), the verification of the regime crossovers of Sec. IV B of the main text in the microscopic parameters, and the conjecture that resource weighting widens spectral gaps are targets for the simulation program accompanying this work.

S9. SKETCH OF AN ENTROPY AND INFORMATION LEDGER

The population ends the cascade holding role structure it did not start with. This section tracks the flows of Fig. 1(a) of the main text (Table S3), marks which generate entropy and which do work with it, and asks whether the first accounts for the second. It is a sketch rather than a derivation, for the reasons the last paragraph gives. Numbers below are computed on a single active axis at the Fig. 1 parameters given in Sec. S10, by `code/entropy_ledger.py`.

a. What the structure is worth in bits. The differential entropy of the ω-identity distribution measures the structure badly, and near onset it moves the wrong way. Two effects compete as Λ_1 rises. The lobes separate, which spreads the distribution, and each lobe narrows as its well steepens. Separation dominates first and narrowing later, so $h(\boldsymbol{\omega} \cdot \hat{\mathbf{g}}_1)$ rises from -0.14 bits at $\Lambda_1 = 0.5$ to 0.74 bits at $\Lambda_1 = 1.3$, then falls through zero to -2.02 bits at $\Lambda_1 = 30$. Just above threshold the differentiated state carries more entropy than the role-free one. Negentropy against a Gaussian of the same variance fails in the other direction, growing without limit to 5.8 bits at $\Lambda_1 = 30$, because it counts the narrowing as well as the separation. Each measures how sharply an ω-identity sits in its well alongside the quantity of interest.

TABLE S3. The flows of the loop, and what each contributes to the ledger. Rates are per agent. $\Delta = \Delta_{im}$ is the status difference read in context m, $S = \Delta + \xi$ the observation through the channel of noise σ_{read}, and $a = \text{sign}(S)$ the action. E_b is the barrier between the two role lobes on an axis.

Flow	Rate	Contribution
Status read, $\boldsymbol{\omega}_j \to S$	νf_m	Acquires $I(a;\Delta)$ bits about the partner's role, against the channel noise σ_{read}
Policy, $a = \text{sign}(S)$	νf_m	Discards $I(S;\Delta) - I(a;\Delta)$ bits of the analog status
ω-identity write, $+b_0 \mathbf{g}_m a$	$\beta = \nu b_0$	Does the work that deepens the wells holding the role label
ω-identity decay, $-\gamma \boldsymbol{\omega}_i$	γ	Relaxes position within a well without erasing the label
ω-identity noise	σ_ω	Thermalizes at $\sigma_\omega^2/2$ and sets the well width and the covariance floor
Barrier escape	$\propto e^{-2E_b/\sigma_\omega^2}$	The one channel that erases a role label
Miscoordination	$\nu f_m q_{im}$	Kicks both ω-identities alike for no payoff, outside any pairwise potential
Resource flow	$\sum_m f_m W_{im}$, c	Converts role information to wealth under the Szilard bound
Schema flow	$\alpha\lambda_m$, $\kappa + \varphi$	Selects the repertoire, which only throughput maintains

The loop creates the label: which lobe an ω-identity occupies on each active axis. That label is one bit per axis at any separation, so the structure is $N(\Lambda_1)$ bits per agent, the counting function of Fig. 2(a) of the main text, capped by the rank of $\mathbf{\Lambda}$. The label also persists, where the rest of the distribution does not. Within a lobe an ω-identity decorrelates in $1/\gamma$, while the label survives until the ω-identity crosses the barrier between lobes. Its height is $E_b = E(0) - E(y^* \hat{\mathbf{g}}_m)$, with E of Eq. (12) of the main text restricted to the line along $\hat{\mathbf{g}}_m$ and y^* the well minimum on it. The ω-identities are distributed as e^{-2E/σ_ω^2}, so a crossing carries the factor $e^{-2E_b/\sigma_\omega^2}$, and $2E_b/\sigma_\omega^2$ rises from 0.6 at $\Lambda_1 = 1.3$ to 50 at $\Lambda_1 = 4$ and 2136 at $\Lambda_1 = 30$. Role structure is the population's broken ergodicity, and by $\Lambda_1 = 4$ a

label's lifetime exceeds every other timescale in the model.

b. What the policy discards. An encounter reads $S = \Delta + \xi$ and emits $a = \mathrm{sign}(S)$. The observation carries $I(S;\Delta)$ bits about the status difference, and the action retains $I(a;\Delta) = 1 - \mathbb{E}_\Delta[H_{\mathrm{bin}}(P_e)]$, the I of the Szilard-form bound in Sec. I C of the main text averaged over the population rather than evaluated at one P_e. Their difference is the information the policy discards, and it is non-monotonic in the drive: 0.010 bits per encounter at $\Lambda_1 = 0.5$, a maximum of 0.126 near $\Lambda_1 = 4$, and 0.009 at $\Lambda_1 = 30$. It vanishes at each end for a different reason. Below threshold the channel carries only 0.027 bits to begin with. Far above it the status distribution has itself collapsed onto two lobes, so at $\Lambda_1 = 30$ the observation carries 0.995 bits and the sign retains 0.985 of them. The coarse-graining comes closest to lossless where the roles are sharpest.

c. The balance. Per encounter the comparison fails, since the policy discards at most 0.126 bits against one bit of role label. Sufficiency comes from the rates. The policy discards $\nu \sum_m f_m [I(S;\Delta) - I(a;\Delta)]$ bits per unit time, while the standing structure needs replacing only at the barrier-escape rate, $N(\Lambda_1)$ bits times $e^{-2E_b/\sigma_\omega^2}$. The ratio is e^{2E_b/σ_ω^2} up to prefactors, which the numbers above put past e^{50} at $\Lambda_1 = 4$. Building the structure costs $N(\Lambda_1)$ bits once, delivered over a transient of many encounters. Entropy production therefore exceeds what the structure requires by a margin exponential in the same drive that creates the structure, and the second law never becomes the active constraint. The other ledger binds instead: the Szilard-form bound of Sec. I C of the main text limits the payoff a bit of role information yields, and at $\Lambda_1 = 30$ it sits within 1.5% of its one-bit maximum. The engine runs against its payoff bound and nowhere near its entropy bound.

d. Efficiency. Three ratios in the ledger are efficiencies, and the drive fixes each of them. The first compares the payoff to the Szilard-form bound it obeys, $\eta_{\mathrm{S}} = W/(2 \ln 2\, w_0 I)$. It depends on the error rate alone, $\eta_{\mathrm{S}} = (1 - 2P_e)^2/(2 \ln 2\, [1 - H_{\mathrm{bin}}(P_e)])$, and runs over $[1/(2 \ln 2), 1]$. The upper end is reached as $P_e \to \frac{1}{2}$, which also fixes the constant in Eq. (3) of the main text, since $2 \ln 2$ is the supremum of $(1 - 2P_e)^2/I$. The bound is therefore tight in the vanishing-signal limit and loose by $1/(2 \ln 2)$ at perfect inference. Across the sweep η_{S} falls from 0.99 at $\Lambda_1 = 0.3$ to 0.72 at $\Lambda_1 = 30$, so the engine converts information to payoff most efficiently where its output vanishes.

The second efficiency is the fraction of the acquired information that survives the policy, $\eta_{\mathrm{P}} = I(a;\Delta)/I(S;\Delta)$, which rises from 0.634 below threshold to 0.99 at $\Lambda_1 = 30$. Its

subcritical value is the hard-decision loss of a Gaussian channel, $2/\pi = 0.637$, which the model would reach if the noise floor left no residual signal, and here leaves a ratio of 0.04.

Their product $\eta = \eta_{\mathrm{S}}\eta_{\mathrm{P}}$ is the payoff obtained per bit the channel delivered, and it is non-monotonic in the drive. The two losses run in opposite directions. Below threshold the policy discards the analog signal while the Szilard conversion stays near-perfect, and deep in the cascade the sign is a near-sufficient statistic while the payoff falls short of the bound. Only near onset are both costly at once, so η passes through a minimum of 0.624 at $\Lambda_1 = 1$, between its limits of $2/\pi$ below and $1/(2\ln 2) = 0.721$ above. The engine reaches its lowest efficiency at ignition.

e. The cost of measurement. Every ratio above assigns no cost to the observation record, and an autonomous engine admits a better treatment. Cocconi and Chen track the hidden self-propulsion state $w = \pm 1$ of a single active particle with an explicit two-state measurement device $s = \pm 1$. The device follows w at rate m with error probability ϵ, while w itself turns over at the tumbling rate α. They define the operational cost of measurement as the excess entropy production rate, the Kullback–Leibler rate between forward and time-reversed path ensembles of the joint process, less what the uncoupled particle produces alone [13].

Their w is the role label $a^* = \mathrm{sign}(\boldsymbol{\omega}_i \cdot \hat{\mathbf{g}}_m)$ here, the persistent variable of the paragraphs above. The pairwise comparison $\mathrm{sign}[(\boldsymbol{\omega}_i - \boldsymbol{\omega}_j) \cdot \hat{\mathbf{g}}_m]$ that one encounter resolves is a different variable, redrawn with every partner, so it is measured once rather than tracked. Their s is the written action a, refreshed at the encounter rate ν, and their closed form then reads

$$\dot{S}_{\mathrm{exc}} = \frac{k_{\mathrm{esc}}\,\nu\,(1-2\epsilon_*)}{2k_{\mathrm{esc}}+\nu}\,\ln\frac{1-\epsilon_*}{\epsilon_*}, \tag{S30}$$

with k_{esc} the rate at which the label turns over and ϵ_* the error of one encounter in reading it.

Both rates need care. A label exists only above threshold, because the subcritical potential has a single well and $\boldsymbol{\omega}_i \cdot \hat{\mathbf{g}}_m$ crosses zero infinitely often, so Eq. (S30) describes the thick phase alone. There k_{esc} is the Kramers rate over that same barrier,

$$k_{\mathrm{esc}} = \frac{\sqrt{|E''(0)|\,E''(y^*)}}{2\pi}\,e^{-2E_b/\sigma_\omega^2}, \tag{S31}$$

Its prefactor climbs from 0.045γ at $\Lambda_1 = 1.2$ to 1.83γ at $\Lambda_1 = 10$, so no fixed rate of the model stands in for it. The exact mean first-passage time matches Eq. (S31) to the factor

of two between reaching the barrier and completing a crossing once $\Lambda_1 \geq 2$, and departs from it fourfold near onset, where the barrier is no larger than $\sigma_\omega^2/2$. The error ϵ_* in turn exceeds the mean-field P_e, because one encounter reads the partner's status alongside the agent's own, a spread the mean field averages away. Computing it is the step this sketch leaves open.

Two statements survive these caveats. Equation (S30) is an entropy rate, so it needs no temperature for σ_ω, and only the payoff-to-energy map stands between it and the efficiency $\langle \dot{W} \rangle / \beta^{-1} \dot{S}_{\rm exc}$ that they define. And the cost falls as $e^{-2E_b/\sigma_\omega^2}$ once the wells are deep, since a label that never turns over needs no second measurement, while it grows without limit as $\epsilon_* \to 0$.

Their controller admits two forms, and Eq. (1) of the main text writes the first. A naive controller acts on the readout as though it were the hidden state, which is the fixed-magnitude kick $b_0 \mathbf{g}_m a$ here. A Bayesian one scales by the posterior, writing $b_0 \mathbf{g}_m \, \mathbb{E}[a^*|S]$ in place of $b_0 \mathbf{g}_m \, a$, the confidence rather than the sign. In their engine the Bayesian form yields more power and an efficiency rising monotonically as measurement degrades, while the naive form reaches its efficiency maximum at intermediate error [13]. The non-monotonic η above is the naive protocol's signature, inverted into a minimum because η omits the cost that Eq. (S30) supplies.

f. Why the loop is not a perpetual motion machine. The coordination income does not come out of the ω-identity noise. The game supplies it and the environment maintains the game. The loop supplies only the role information that lets a pair reach the payoff above Nash, and $W \leq 2 \ln 2 \, w_0 I$ caps that conversion at one bit per encounter, with equality requiring $\sigma_{\rm read} \to 0$. The loop also cannot amplify itself without limit. Raising the payoff requires lowering P_e, hence a larger Δ_{im}. The saturating score caps $|\Delta_{im}|$ at 2 however large the ω-identities grow, and the schemas that raise it are limited by the shared budget φ. The fixed point of Sec. II B of the main text is where those limits meet the payoff inflow.

g. How far this is an accounting. Three gaps separate the ledger above from a derivation. First, the two informations are not conjugate. $I(S;\Delta) - I(a;\Delta)$ concerns the partner's continuous status, while the role label concerns the agent's own persistent position, so showing that the first accounts for the second rather than merely exceeding it requires the encounter-level treatment the mean field removes. Second, the reduced dynamics shows none of this on its own. At fixed schemas the drift is the gradient of Eq. (12) of the main text with

a Boltzmann stationary distribution, which produces no entropy and does no work. The drive has been absorbed into E, and the resulting Boltzmann form is the stationary state of a driven process rather than an equilibrium. Third, the model carries three scales and no energy units. The ω-identity temperature $\sigma_\omega^2/2$ has units of $|\boldsymbol{\omega}|^2$ per unit time, because E is the potential of a velocity. σ_{read} is a status difference, and w_0 is a payoff. Writing the ledger in one unit needs two ratios among them, and the model fixes neither.

One ratio has a standard resolution. A binary choice made through a noisy observation takes the functional form of a quantal response, and $\bar{\pi}(z) = \Phi(z/\sigma_{\text{read}})$ is the probit counterpart of the logit whose scale parameter is an inverse rationality [14]. That same scale is the temperature of entropy-regularized control, where the value function becomes a free energy [15, 16]. Reading σ_{read} as that temperature converts payoff to nats in the choice sector. The correspondence is one of functional form alone, because the agents here read a status signal and maximize nothing. The schema sector admits no such reading, because Eq. (4) of the main text carries no noise, and a finite population would be needed to supply one with the population size as the inverse temperature [17].

The ratio left open is the one between the ω-identity bath $\sigma_\omega^2/2$ and the choice temperature σ_{read}. Two noise scales at different effective temperatures on coupled degrees of freedom are the generic setting for a work-producing steady state, so the loop may draw its work from that mismatch. We do not establish it here, because σ_{read} is measurement noise refreshed at each encounter and not a reservoir with its own dynamics. Until the ratio is fixed the bound of Sec. I of the main text stays in information units, and Ref. [18] sets out the form such an inequality takes once bits and work share a scale.

A full accounting would also differ in form. The loop is autonomous rather than cyclic, so no measurement-and-reset cycle exists to carry it. The information-flow formalism for coupled subsystems [19] is the standard replacement, and it requires bipartite dynamics, in which the two subsystems never update in the same instant. This model is not bipartite, because a miscoordinated encounter kicks both partners' ω-identities at once. The excess entropy production of Eq. (S30) carries no such requirement, which is why the paragraph above takes that route [13]. Refs. [7, 20] carry the accounting out for a single agent operating a cyclic engine. The population case differs again in that both members of an encounter measure and are measured, so each serves as the other's memory and no subsystem plays the demon alone.

S10. SIMULATION DETAILS

$N_a = 2500$ agents, $d = M = 3$, $\gamma = 1$, $\sigma_{\rm read} = 0.6$, $\sigma_\omega = 0.25$, Euler–Maruyama integration of Eq. (1) of the main text at step $dt = 0.02$, in its dense-encounter limit. At each step every agent meets $K = 32$ partners drawn at random in every context ($f_m = 1$). Its single-sample sign decision is replaced by the noise-averaged action of Appendix A of the main text at $T = 1$, and the average over the K partners enters the drift with weight β. The write therefore carries no shot noise, and the only noise is the isotropic σ_ω, with the flat floor $\sigma_\omega^2/2\gamma = 0.031$. Finite K leaves a residual sampling noise. At $\beta = 0.1$ the variance along each axis is 0.032–0.033.

The schema repertoire is held fixed, with orthogonal schema axes and a hierarchical profile of strengths ($|\mathbf{g}_m| = 0.533$, 0.434, 0.302), so the covariance eigenvalues $v_k(\beta)$ resolve the cascade at $\Lambda\mu_k = 1$ with $\mu_k = u_k = |\mathbf{g}_k|^2$, the eigenvalues of $\boldsymbol{H}^\top\boldsymbol{H}$, and $\rho_{\rm pair} = 0$ (factor 1) for random pairing. The thresholds are $\beta_c^{(k)} = 2.65$, 3.99 and 8.25. The protocol is quasi-static: at each β on a grid we equilibrate (1.2×10^3 burn-in steps, warm-started from the previous β) and average the order parameter over a further 1.5×10^3 steps, so panels b,c are functions of β rather than of time. The endogenous selection of the cascade ordering under dynamic schemas is treated in Sec. IV A of the main text. The full dynamic-schema code is available in the repository given under Data availability in the main text.

The learning simulations of Sec. S1 D use the encounter process at finite rate instead. Each step is one encounter round: one random pairwise matching, all M contexts played per encounter at $f_m = 1$, and each agent's single-sample sign decision drawn and written with weight $\beta\, dt$. The rate is $\nu = 1/dt = 50$. The flat noise floor requires $\beta^2/\nu \ll \sigma_\omega^2$, here $\beta \ll 1.8$, so those runs are in the shot-noise regime of Appendix A of the main text. There the thresholds $\Lambda_k = 1$ are unchanged and the pre-onset amplitudes exceed the flat floor. With the policy slope set to zero (Sec. S1 D), so that only the shot noise of the write acts, the variance along axis 1 is 0.043, 0.135 and 0.447 at $\beta = 2$, 6, 12, against the flat floor 0.031.

---

[1] B. VanSaders, M. Fruchart, and V. Vitelli, Measurement-induced phase transitions in informational active matter, PNAS Nexus **5**, 10.1093/pnasnexus/pgag077 (2026), arXiv:2302.07402

[cond-mat.stat-mech].

[2] T. Sagawa and M. Ueda, Generalized Jarzynski equality under nonequilibrium feedback control, Phys. Rev. Lett. **104**, 090602 (2010).

[3] R. J. Aumann, Subjectivity and correlation in randomized strategies, Journal of Mathematical Economics **1**, 67 (1974).

[4] R. J. Aumann, Correlated equilibrium as an expression of bayesian rationality, Econometrica **55**, 1 (1987).

[5] M. Puelma Touzel, The fitness landscape of social norms in social dilemmas (2026), arXiv:2605.18834 [cs.GT].

[6] D. Bergemann and S. Morris, Bayes correlated equilibrium and the comparison of information structures in games, Theoretical Economics **11**, 487 (2016).

[7] D. Daimer and S. Still, Thermodynamically rational decision making under uncertainty (2023), arXiv:2309.10476 [physics.data-an].

[8] R. Selten, A note on evolutionarily stable strategies in asymmetric animal conflicts, Journal of Theoretical Biology **84**, 93 (1980).

[9] J. Maynard Smith and G. A. Parker, The logic of asymmetric contests, Animal Behaviour **24**, 159 (1976).

[10] J. Maynard Smith, *Evolution and the Theory of Games* (Cambridge University Press, Cambridge, UK, 1982).

[11] A. P. Castro, J. Bechhoefer, and D. A. Sivak, Harnessing higher-dimensional fluctuations in an information engine, Phys. Rev. Lett. **136**, 047101 (2026).

[12] D. Krotov and J. J. Hopfield, Dense associative memory for pattern recognition, in *Proceedings of the 30th International Conference on Neural Information Processing Systems*, NIPS'16 (Curran Associates Inc., Red Hook, NY, USA, 2016) p. 1180–1188.

[13] L. Cocconi and L. Chen, Efficiency of an autonomous, dynamic information engine operating on a single active particle, Phys. Rev. E **110**, 014602 (2024).

[14] R. D. McKelvey and T. R. Palfrey, Quantal response equilibria for normal form games, Games Econ. Behav. **10**, 6 (1995).

[15] H. J. Kappen, Linear theory for control of nonlinear stochastic systems, Phys. Rev. Lett. **95**, 200201 (2005).

[16] S. Levine, Reinforcement learning and control as probabilistic inference: Tutorial and review,

arXiv:1805.00909 10.48550/arXiv.1805.00909 (2018).

[17] G. Sella and A. E. Hirsh, The application of statistical physics to evolutionary biology, Proc. Natl. Acad. Sci. U.S.A. **102**, 9541 (2005).

[18] S. Still, D. A. Sivak, A. J. Bell, and G. E. Crooks, Thermodynamics of prediction, Phys. Rev. Lett. **109**, 120604 (2012).

[19] J. M. Horowitz and M. Esposito, Thermodynamics with continuous information flow, Phys. Rev. X **4**, 031015 (2014).

[20] S. Still, Thermodynamic cost and benefit of memory, Phys. Rev. Lett. **124**, 050601 (2020).